

Beyond the First Battle: Navigating Uncertainty in War's Later Stages

Zhejun Qiu* Perisa Davutoglu[†] William Spaniel[‡]

August 15, 2026

Abstract

Many wars are fought over distinct phases, where some underlying capabilities in one portion may have little or no bearing on others. However, existing models of bargaining while fighting treat battles as identical. We relax that assumption and examine what happens when a source of uncertainty only affects later phases. Under such conditions, learning becomes more difficult. More specifically, no separating equilibria exist if types have identical likelihoods of winning the upcoming battle, even if later battles do depend on type. Thus, we have a natural information-based explanation for prolonged fighting. A case study of the Shining Path in Peru illuminates the mechanism.

*PhD Candidate, Department of Political Science, University of Pittsburgh.

[†]PhD Candidate, Department of Political Science, University of Pittsburgh.

[‡]Associate Professor, Department of Political Science, University of Pittsburgh.

1 Introduction

If war is the result of private information, then a conflict will end when the process of fighting and bargaining has revealed enough information to resolve the problem. This is the central conclusion of a large literature that treats war as a process (Filson and Werner 2002; Slantchev 2003; Powell 2004). In particular, an uncertain state can structure its proposals to induce types with lower war payoffs to accept in earlier rounds of negotiations. The act of rejecting and fighting credibly screens types. In turn, strong types accept later proposals. This process ensures that wars as a consequence of uncertainty may not continue indefinitely.

A central assumption in these models is that private information has similar effects across each battle—e.g., a given type pays the same cost of war in each round or wins each battle with the same probability. This is a sensible modeling choice for a first cut at the problem, as it provides an ideal “experiment” for a model of learning (Paine and Tyson 2020). It is also substantively plausible if the source of uncertainty is over a state’s resolve or its general martial effectiveness.

However, a source of uncertainty may only affect war outcomes for particular phases of the conflict. For example, one source of uncertainty in the Russia-Ukraine War is Western commitment to Kyiv *in the long run*. Results from initial periods of conflict are not directly informative of that. Ukraine’s ability to withstand Russia’s initial assault might not give Moscow a clearer picture of whether Western assistance will flow years down the line. More explicitly, the Kremlin concluded that its initial blitz would render the West helpless to provide aid but had uncertainty over how much aid the West would provide conditional on survival. What happened during the first phase of the conflict therefore did not provide direct information about what would happen afterward.

This situation is not unique. Other questions arise with clear geographic divides in the battle space. Some wars begin in rural areas but culminate in urban centers. This is relevant because military tactics in one domain may not carry over to another. For example, the Iran-Iraq War began in the plains of Iran’s Khuzestan province, but Iraq’s mechanized brigades faded into the background as the battle shifted into the Zagros Mountains. Similarly, in the Pacific War, the United States had to undergo preliminary naval battles before hopping between remote islands. On the eve of World War I, leaders understood that there would be two stages: an initial and hopefully decisive

confrontation with existing military capacity and a later war of attrition otherwise (Philpott 2015, 42–43). The latter, from the British perspective, would last “just as long as the Prussian militarists can impose their will upon a suffering nation” (Philpott 2015, 95), the answer to which was subject to debate within Britain (Philpott 2015, 71) and unanswerable until the peacetime capabilities dwindled. Many modern conflicts begin with a period of aerial bombing before an invading army takes over. Some wars—for example, the 2001–2021 War in Afghanistan—further transition into a third phase of insurgency.

The issue also arises outside of shifting combat domains. If the underlying question is about a country’s ability to militarily innovate over the course of the war, then expectations regarding initial outcomes are identical across types. Likewise, if there is doubt about an invader’s ability to handle long-distance logistics within conquered territory, the true tests must occur after territory has changed hands. If the question is about whether a government can mobilize for a larger campaign after initial setbacks, then the initial setbacks need to happen before that is observed. Electoral events with fixed calendar dates—and war efforts that depend on the outcomes—fit here as well. For example, if the parties disagree over the relative probabilities of a presidential election and expect assistance would vary under the two administrations, then initial outcomes do not provide insight into what could happen after the election.

It is not immediately clear how learning works under these conditions, especially when the source of uncertainty is in the war’s latter phases. At one extreme, one might suspect that the fear of reaching unfavorable segments might still compel weaker types to accept earlier proposals. Calculations of various equilibrium cutpoints might change, but the overall lessons from the bargaining and learning literature would remain the same. At the other extreme, one might worry that *no* learning takes place in periods where types are mechanically identical. The temptation to bluff is too strong, and no proposal strategy is capable of screening away weaker types.

In search of answers, we develop a crisis bargaining model where types have identical likelihoods of succeeding initially but one type is stronger in the latter phase. Like standard models (Filson and Werner 2002), the uncertain party makes an initial proposal, which the informed party accepts or rejects. If it accepts, the interaction ends with the proposed distribution. If it rejects, the parties fight a battle. If they survive, then they play another ultimatum game, which either ends in a settlement or a final

phase of the war.

Interestingly, neither of the earlier perspectives is fully correct. The headline result, however, is bad news. When war breaks out with late-stage uncertainty, the expected duration of conflict is longer when compared to uniform uncertainty or early-stage uncertainty. The intuition is straightforward. It is not possible for a weaker type to fully separate in the early phase of the war. Such a hypothetical equilibrium would require that the various types face differential costs or risks for the upcoming battle. This is not the case in our model with late-stage uncertainty but when uncertainty is uniform across battles. Hence we cannot recover separating equilibria, whereas models like Slantchev (2003) can.

The good news, contrary to the pessimistic expectation, is that *some* learning can take place. When the proposer wishes to make a risky demand, it selects the equivalent of the weak type’s war payoff in the first stage. The weak type responds by mixing, sometimes accepting that proposal and sometimes rejecting. Thus, some level of screening occurs, with the weak type more likely to accept. Moreover, peace prevails in the first period with positive probability in all equilibria. Unfortunately, though, the learning here is weak. If the parties reach the later stage, the equilibrium proposal always induces the weak type to accept and the strong type to fight. In other words, the information gleaned is not particularly useful.

Our main substantive takeaway is that we have an information-based explanation for prolonged fighting. Powell (2006, 169) states that “informational explanations and the models underlying them have [a] major [limitation]: they often provide a poor account of prolonged conflict.” We are not the first to show that lengthy delays are possible. In Slantchev (2003) and Powell (2004), for example, the the uniformed actor may adopt a “skimming” strategy, where it sequentially buys off progressively tougher types after each round of fighting. Wars can last relatively long times because the number of types may be many. Nonetheless, each round of fighting perfectly screens out a type. This is not possible in our model. Similarly, in Smith and Spaniel (2019), unproductive learning can occur if the proposer’s prior belief indicates that it is very likely facing the weakest type, but clean screening remains possible for more moderate priors. However, we show that moderate priors generate unproductive learning with late-stage uncertainty.

This finding helps speak to an ongoing debate on war duration. The long length

of the average civil war often serves as evidence for why commitment problems are better suited for explaining intrastate conflict (Walter 1997; Fearon 2004). Commitment problems clearly do explain a central logic of many civil wars. However, the results from our model indicate that the length of a conflict by itself may not be a good indicator of the information/commitment divide.

We take two additional steps. First, in an extension, we generalize the model to allow for each party to have more units and therefore survive through an arbitrarily long series of battles. For a given period, full separation can only occur if the forthcoming battle poses a differential risk on the party. Thus, poor information transmission is sustainable for an arbitrarily large number of initial battles if outcomes are not a function of the type. Whether informational narratives are good fits for civil wars therefore depends on *when* the private information impacts the battle space.

Our second step reinforces this assertion. We explore Peru's war against the Shining Path in the 1980s and 1990s. A central problem with the conflict was how the Shining Path would fare in urban battles, which was necessarily the second phase of a war. The government, given its geographic positioning, had private and optimistic beliefs about its chances there. Consistent with our model's prediction, the first phase of the war did not provide meaningful information to the Shining Path. As a result, it continued into the second phase.

We also uncover a few additional formal results. First, we show that the efficiency implications of long-term uncertainty are mixed. Precisely because screening is more difficult under these circumstances, the situations where war occurs at all are smaller here than in comparable models. Second, we switch from uncertainty over the probability of victory to uncertainty over costs. The headline result is the same as before. Third, we flip the timing so that the type only matters for the initial battle. The welfare implications are mixed. Obtaining a peaceful result from the start is harder when uncertainty only matters in the first stage. However, conditional on war occurring in the first stage, settlement is guaranteed in the second. Note that this is not the consequence of a standard signaling process. Rather, the information problem disappears after the first battle by default. Finally, we show that the lack of separating equilibria extend to environments where the first battle is semi-informative.

2 The Model

Our main task is to explore the consequences of late-stage uncertainty in crisis bargaining. Per Paine and Tyson (2020), the best way to do this is conduct an “experiment” by holding constant an existing framework and swapping out that single assumption. We do this below, using Filson and Werner (2002) as our signpost. Later, we cover richer (and longer) bargaining environments.

Two actors, A and B, negotiate how to divide a good with a value standardized to 1. Following the most common way to incorporate bargaining while fighting in the literature (Filson and Werner 2002; Spaniel and Bils 2018; Smith and Spaniel 2019; Krainin, Thomas and Wiseman 2020), we adopt a unit approach, where each side has a fixed number of units, and losing a battle eliminates one unit. The parties can continue until they run out of units.

In the interest of establishing the main substantive finding, we wish to examine the simplest extensive form that generates the key result. Fortunately, we can show the absence of clean separation in a game where A has one unit and B has two. Three total units is the minimum to allow for multiple rounds of bargaining, as one party must have a spare unit for a battle to not be game ending. Later, we should that the key result—the absence of a separating equilibrium—is endemic no matter how many units the parties have.

Nature begins the game by drawing B as a weak type with probability $q \in (0, 1)$ and as a strong type with probability $1 - q$. Bargaining takes place over two stages, and the game can end via settlement or war. In the first stage, A proposes $x_1 \in [0, 1]$. B sees the proposal and accepts or rejects. If B accepts, the game ends with that division. If it rejects, the parties fight a battle. With probability $p_1 \in [0, 1]$, A wins, and bargaining advances into the second stage. With probability $1 - p_1$, B wins, thereby eliminating A and ending the game.

If the parties reach the second stage, A makes a new demand $x_2 \in [0, 1]$. B again accepts or rejects. Accepting ends the interaction with the proposed division. Rejecting leads to another battle. Unlike the first stage, the probability of victory depends on B’s type. In particular, if Nature drew the weak type, A prevails with probability $p'_2 \in (0, 1]$; if Nature drew the strong type, A prevails with probability $p_2 \in [0, p'_2)$. Note that this probability references an earlier draw from Nature, meaning that B knows what this

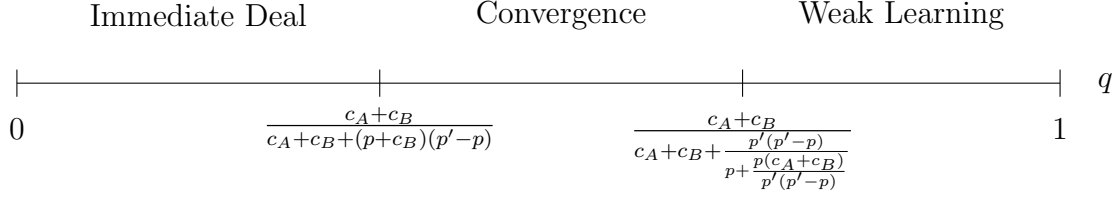


Figure 1: Equilibrium predictions of A's belief q in standard bargaining-while-fighting models.

probability will be as it progresses through the earlier stage.¹

Payoffs are as follows. If the parties settle in the first stage, each side receives its corresponding share. Thus, A earns x_1 and B earns $1 - x_1$. Likewise, if the parties agree in the second stage, the settlement payoffs are x_2 and $1 - x_2$, respectively. However, fighting battles is costly: A pays $c_A > 0$ for each one, while B pays $c_B > 0$. Thus, the total payoffs for a settlement in the second stage—conditional on reaching that point—are $x_2 - c_A$ and $1 - x_2 - c_B$. If one side runs out of units, it receives none of the good, while the other side takes all of it. They must also pay however many costs have accumulated over time. To avoid corner solutions that do not yield additional theoretical insight, we assume that the weak type of B has a positive war payoff.

2.1 Predictions of Existing Bargaining-while-Fighting Models

As a starting point for our equilibrium results, it is worth briefly pausing to describe how traditional bargaining-while-fighting models work for analogous setups. More explicitly, suppose that B's type governed the outcomes of both battles. Figure 1 summarizes equilibrium predictions as a function of A's prior belief about B, using p' to represent a more powerful type's probability of winning in every period and p to represent a weaker type's probability of winning.²

In the leftmost region, A believes B is very likely the strong type. Attempting to use war as a means to screen types is a losing effort for A. Indeed, the strong type is so frequent that trying to withhold concessions results in fighting the first battle

¹Another way to model late stage uncertainty would be to draw (or redraw) B's type before each battle. It would be less interesting if learning failed here, because B cannot signal in the first stage something it does not know about the second stage.

²See Appendix A.5 for derivation of this, substituting the special case.

against those strong types. Moreover, the weak type is so infrequent that if it just bluffed strength by pooling with the strong type on rejection, A would still want to pay out concessions as though it were facing the strong type. That whole process is costly, though, and A prefers to avoid suffering that disutility. As such, it offers B an amount commensurate with the strong type’s war payoff, matching what happens in the extreme case when q goes to 0 and the game moves to complete information.

The middle region reflects the traditional learning result. The temptation for a weak type of B is to reject the first stage proposal, attempt to bluff strength by emerging victorious in the battle, and then receive a proposal intended to buy off the strong type in the second stage. However, A can remove that temptation by offering a premium. This proposal falls in between the weak type and strong type’s war payoff. Specifically, A discounts the concessions it would make to the strong type by the extra risk the weak type incurs by attempting the bluff. This is better for A than trying to drive a hard bargain because the weak type is not particularly frequent. Clean separation therefore occurs, allowing A to update that B is strong in the second stage and settle with that type and avoid completing the war. Note further that it is *not* the battlefield results that influence A’s belief in the second stage. Rather, the bargaining strategy that A pursues incentivizes separation. As a result, even though in our model the first battle functions identically for each type, it is not obvious that separation is doomed for failure.

Nevertheless, the separating logic falls apart in the rightmost region. Clean separation requires that A pay a premium to the weak type. As the weak type becomes almost certain, it no longer finds paying that premium worthwhile. Instead, A holds firm to offering a small quantity equivalent to the weak type’s war payoff. The weak type then mixes between accepting and rejecting in the first stage as a means to prevent A from learning for free. What happens afterward depends on the source of uncertainty. When A is uncertain over B’s per-period cost (which we explore in an extension), the low cost type ultimately accepts in the second stage while the high cost type still rejects. Hence, “weak” learning occurs here—A’s second-stage posterior does not equal its first-period prior, but the new information does not meaningfully change its proposal strategy. When the source of uncertainty is the probability of victory in battle, then A must update its belief both from the rejected proposal and the result of the battle. Regardless, the key point is that the first-stage separation fails.

Below, we will see that when uncertainty is restricted to the second period battle, the convergence parameter space collapses. Learning still occurs under these circumstances. However, it is unhelpful in the sense that the weak type of B semi-separates, allowing A's information problem persists into the second period.

2.2 Equilibrium

Before delving into what happens in the first stage, exploring the incentives in the second stage will prove helpful. Whatever signals come from the first stage, A forms a posterior belief about B's type for the final battle. It therefore faces a traditional risk-return tradeoff. Demanding a large quantity has the benefit of extracting an attractive settlement if B is weak and accepts. However, it also comes at the risk of the strong type rejecting. The alternative is to make a generous concession to ensure the strong type's compliance. This saves A the risk of paying its own cost and guarantees extracting at least some of the surplus out of B. The downside is that the extracted surplus is less than in the previous case.

Let r be A's posterior belief that B is the weak type in the second stage. The appendix shows that A prefers taking the risk if:

$$r > \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \quad (1)$$

Analogously, A prefers making the conservative demand when the inequality is flipped and is indifferent when it is equal.

Per the earlier discussion of what occurs in traditional bargaining-while-fighting models, we might predict there will be three types of equilibrium behavior based on A's prior belief: (1) very conservative demands that guarantee acceptance in the first stage, (2) moderate demands in the first stage that separate the weak type in the first stage and lead to generous offers to the strong type in the second stage, and (3) aggressive demands throughout. Our first main result is that this is no longer the case:

Proposition 1. *No separating equilibrium exists.*

This result stems from the intuition we developed earlier. The reason separation works in standard models is that fighting the first battle is differentially costly or risky to the inferior type. The proposer exploits that by taking what the superior type would

receive from a settlement in the second stage and then applying the differential cost or risk of the first stage. The resulting first-stage proposal therefore equals the best the receiver could hope to earn from fighting but is still less than what the superior type receives via war. Hence separation is feasible.

That logic falls apart here. Imagine the types separated in the first stage. Then A will form a posterior belief at the onset of the second stage that B is certainly the strong type. Based on that, A's concession will be commensurate with B's stronger ability to win the second battle. Notably, the transfer occurs through peaceful means because A wants to extract the surplus that comes with avoiding further conflict.

However, the weak type has no incentive to accept an initial proposal that the strong type would refuse. Again, the types face no differential cost or risk to the first battle. Thus, imagine that it is profitable for the strong type to reject in the first stage. This implies that given a p_1 chance that A wins the first battle and the c_B cost it pays are worth suffering to convince A to make a generous concession in the second period. But the weak type also can take the p_1 chance with the c_B cost, fool A into proposing the generous concession, and enjoy a better deal than it otherwise receives by accepting the first proposal. Consequently, separation fails.

The expectation is similar to the central finding in Spaniel and Bils (2018), but the mechanism is distinct. They also construct a bargaining-while-fighting game that fails to produce a separating equilibrium. However, their source of uncertainty is the extent to which one party wishes to conquer the other. For interior solutions, the proposal necessary to buy the compliance of the more expansionist type only includes parts of the pie that the less expansionist type wants to capture. As a result, first-stage separation fails.

The subtle difference between the mechanisms is that the types in Spaniel and Bils (2018) never face differential risks or costs throughout the interaction. Rather, the difference in payoffs only arises in what each type chooses to do post-victory. Mathematically, though, in both models, the types do not have any distinct tradeoffs until the second round of bargaining. Viewing this in terms of a costly signaling framework, this is why other models yield separating equilibria but our model and the model in Spaniel and Bils (2018) do not.

Those comparisons aside, we are still left to derive our actual equilibrium. The key initial step to solving the game is to make two inferences about proposals that B must

either accept or reject. Because A will never propose more than what is necessary to induce the strong type to accept in the second stage, we can establish an upper bound on demands that both types must accept. Specifically, the appendix shows that both types must pool on accepting in the first stage for any:

$$x_1 \leq p_1(p_2 + c_B) + c_B \quad (2)$$

The remainder of this value is the strong type's payoff for a war through both stages; this is why it uses p_2 and not p'_2 . The strong type is willing to accept because rejecting yields the same payoff. Meanwhile, the weak type has an even greater incentive to accept because it earns strictly more than what it could obtain through war. Furthermore, trying to bluff strength in the first period by rejecting does not work because the strong type is unwilling to do the same.

Similarly, we can establish a lower bound on when both types must reject. Specifically, the appendix shows that war occurs in the first stage with probability 1 for any:

$$x_1 > p_1(p'_2 + c_B) + c_B \quad (3)$$

Note that the difference between Lines 2 and 3 are flip-flopping p_2 and p'_2 . Line 3 uses p'_2 because the remainder of it is the weak type's payoff for a war through both stages. If A demands an amount greater than that, the weak type would never accept; it could deviate to fighting in both stages and guarantee a greater payoff. Obviously, if the weak type can do better through war, so too can the strong type. As such, the strong type must also reject the such proposals.

The bulk of the work in the appendix is dedicated to explaining what happens in the regions between Lines 2 and 3. As Proposition 1 established, separation is not possible. This gives rise to the incentive to deviate that Proposition 1 explored. In turn, the only option left for the weak type is to semi-separate.

With all of that out of the way, we are ready to describe the equilibrium. Only two parameter spaces exist, with

$$q^* \equiv \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \right) \left(\frac{c_A + c_B + (1 + p_1)(p'_2 - p_2)}{c_A + c_B + p_1(p'_2 - p_2)} \right) \quad (4)$$

serving as the cutpoint that lies between them. A's main decision is between two demands. The first is $x_1 = p_1(p_2 + c_B) + c_B$, which leaves an equivalent for B equal to the strong type's war payoff. Henceforth we refer to this as a "conservative demand." The second is $x_1 = p_1(p'_2 + c_B) + c_B$, which leaves an equivalent for B equal to the weak type's war payoff. Henceforth we refer to this as an "aggressive demand." Below, we see that the cutpoint on q determines A's decision.³

Proposition 2. *Suppose A's prior belief that B is weak is sufficiently low (i.e., $q < q^*$). Then A makes a conservative demand in the first stage. Both types pool by accepting.*

Proposition 3. *Suppose A's prior belief that B is weak is sufficiently high (i.e., $q > q^*$). Then A makes an aggressive demand in the first stage. The strong type always rejects, while the weak type mixes between accepting and rejecting. In the second stage, A demands an amount only the weak type accepts, while the strong type fights a second battle.*

In essence, A still faces a risk-return tradeoff, similar as it would in a one-period game. The added complication is trying to figure out how to handle the weak type's incentive to bluff strength, given that it cannot use the normal screening offer to separate types. The appendix shows that A can only pursue bad solutions to this problem. First, A would never propose $x_1 < p_1(p_2 + c_B) + c_B$. Per the discussion around Line 2, $x_1 = p_1(p_2 + c_B) + c_B$ is sufficient to induce both types to accept, so demanding anything less makes an unnecessary concession. Second, A would never propose $x_1 > p_1(p'_2 + c_B) + c_B$. Per the discussion around Line 3, that guarantees both types reject. Consequently, A could do better by ensuring that it secures some of the surplus with a non-zero probability in the first stage.

More subtly, the appendix demonstrates that A's payoff for any proposal $x_1 \in (p_1(p_2 + c_B) + c_B, p_1(p'_2 + c_B) + c_B]$ is strictly increasing. These are the quantities that induce the weak type to semi-separate. The weak type's goal with this strategy is to leave A unsure what to do when it reaches the second stage. This requires a posterior that equals Line 1's right hand side. Note that this is *not* a function of A's proposal and thus A's specific demand within this region does not influence the probability of a

³The complete description of two equilibria, including the off-the-path decisions and a summary table, is provided in Appendix Section A.1.2.

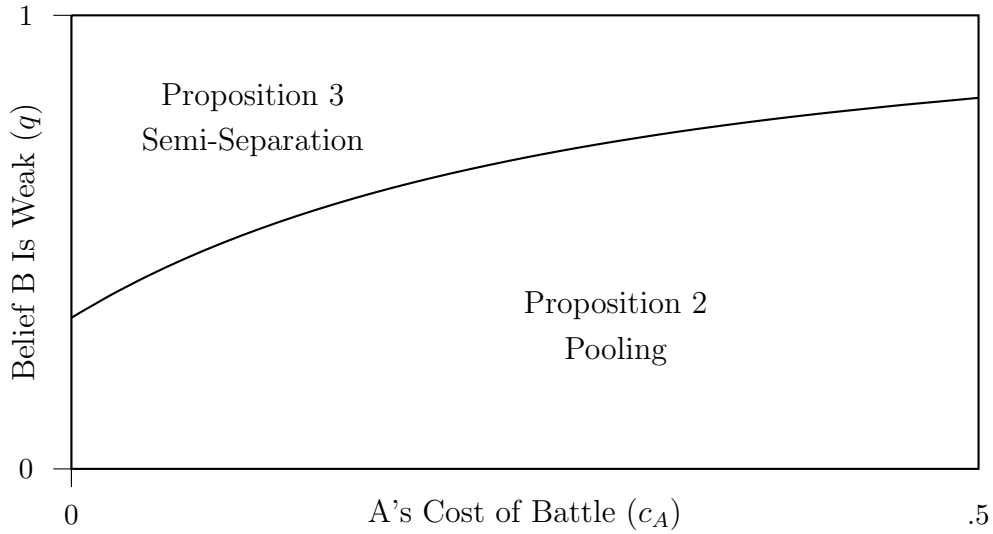


Figure 2: Equilibrium plot of the main model. The line demarcating the two parameter spaces is q^* , as defined on Line 4.

first-period battle. In turn, if A wants to pursue semi-separation, the best it can do is demand the largest quantity in the region, which is $x_1 = p_1(p'_2 + c_B) + c_B$.

Zooming out, Propositions 2 and 3 address the risk-return tradeoff that A faces. When A's prior belief that B is weak is sufficiently low, then trying to sort through the semi-separating behavior is not worthwhile. Instead, A demands $x_1 = p_1(p_2 + c_B) + c_B$. This is the amount commensurate with the strong type's war payoff and thus both accept. But when A's prior belief that B is weak is sufficiently high, then the potential return justifies the headaches that come along with inducing semi-separation.

3 Substantive Implications and a Theory of Prolonged Wars

To help illustrate the logic of the equilibrium, Figure 2 plots our two parameter spaces as a function of c_A and q . The divider is Line 4. When q is small and c_A is large, Proposition 2's logic wins out. This is because A is sufficiently convinced that B is the strong type, and its costs of trying to execute a noisy screening of types is large. It therefore settles on making a generous proposal to induce both the types to accept.

In contrast, when q is large and c_A is small, the incentives are flipped. Indeed, our model provides a clear explanation for prolonged wars as a result of information problems:

Remark 1. *Across equilibrium parameter spaces, if war occurs in the first stage, then it must occur with positive probability in the second stage.*

This is a consequence of the conditions for a semi-separating equilibrium. A's first-period proposal leaves the weak type indifferent between accepting and rejecting. To maintain the indifference condition, A cannot provide *any* additional concessions in the second stage. Thus, the second-stage proposal must induce the strong type to reject. As a result, if a late-stage information problem causes a war to occur, then it must continue to the end some portion of the time. In other words, and in contrast to the conventional wisdom from Powell (2006), we have an information-based explanation for prolonged wars.

It is worth noting that wars can continue until the end within the standard bargaining-while-fighting setups. However, the existence of separating equilibria in such models clouds the prevalence of such outcomes. Our point here is that the source of uncertainty matters is making empirical predictions about the length of wars. For example, any concerns about long-term Western support for Ukraine on the eve of Russia's invasion should have made analysts uneasy about the expected duration of the conflict. In contrast, something like general martial effectiveness (Arena 2013) as a cause for conflict initiation should be less concerning.

To further underscore the importance of modeling these types of questions, Figure 3 illustrates a counterintuitive implication regarding the inefficiency of war. Previously, Figure 2 showed the standard monotonic relationship between the cost of war and incentives to screen. That is, as the costs increased, A's incentive to make risky demands decreased. However, that is not the full story. Conditional on A preferring to take a risk, the probability of war *increases* as costs increase.

What accounts for this strange discrepancy? Suppose the parameters fall into Proposition 3's semi-separating equilibrium. Unlike a one-shot ultimatum game, mixing strategies determine the outcome of the first stage. In particular, the probability that a war breaks out is the probability that A is facing the strong type, plus the probability that A is facing the weak type multiplied by the probability the weak type rejects.

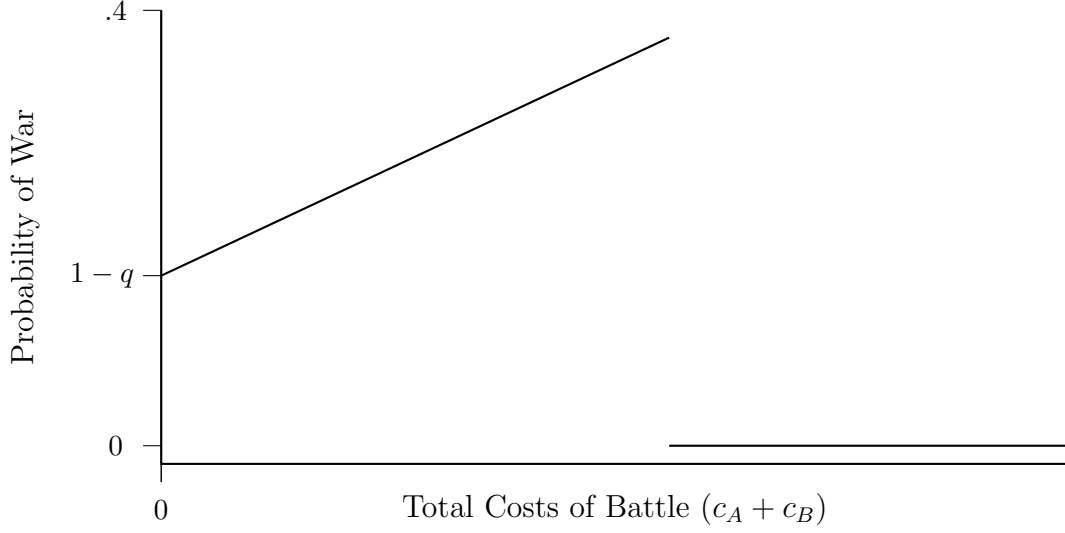


Figure 3: The Total Inefficiency of Battles ($c_A + c_B$) and the Probability of War

Substituting the equilibrium probability of the weak type rejecting that is calculated in the appendix, this probability is:

$$1 - q + q \left[\left(\frac{1 - q}{q} \right) \left(\frac{c_A + c_B}{p'_2 - p_2} \right) \right] = (1 - q) \left(1 + \frac{c_A + c_B}{p'_2 - p_2} \right)$$

By simple inspection, it is clear that the probability of war increases in the social costs of war, and moreover that it is entirely through the weak type's higher probability of rejection. To understand why, recall that the goal of the weak type is to make A unsure about what to do in the second stage. As the cost of war increases, A becomes *less* inclined to make risky demands in the second stage, as extracting a larger surplus sounds more attractive. The weak type's mixed strategy must counteract that incentive. To do so, the weak type must reject more often so that making risky demands in the second stage remain attractive to A. But that means war breaks out more often.

Overall, Figure 3 shows that the relationship is nonmonotonic. When the costs of battle are small, A is incentivized to try to screen B's types. Thus, it chooses to make the risky demand. However, consistent with the logic just described, the weak type rejects more often as the costs increase. Eventually the costs become so large, though, that A switches its strategy to the generous first-period proposal. As a result, the probability of war drops to zero.

4 Extensions

Our main substantive question is what happens in models where battles are not necessarily uniform. Our main result is that uncertainty over late-stage battle outcomes prevents clean separating equilibria. This raises a few natural questions. First, how problematic is late-stage uncertainty if the interaction could last more than two periods? Second, is this result unique to the ability to win battles? Third, how do the results compare to situations where initial battles entail uncertainty but later battles do not? And fourth, how sensitive are our results to there being *no* difference in first period capacity?

We answer each of those questions in this section. Broadly, we find that late-stage uncertainty is pernicious, is not sensitive to the source of the uncertainty, and has ambiguous welfare implications compared to initial uncertainty.

4.1 Generalized Battlespace and Long Wars

To begin, one might wonder whether this result extends past the simple two-period game that the main model developed and just how prolonged a war could be if there were more periods. In the appendix, we develop a richer game. In it, A begins with $\alpha \in \mathbb{N}$ units and B begins with $\beta \in \mathbb{N}$ units, both publicly known. Nature still begins in the interaction by drawing B as a weak type θ_W or as a strong type θ_S . Each period, A makes a proposal $x_t \in [0, 1]$, which B accepts or rejects. Accepting ends the game. Rejecting leads to a battle, which costs c_A and c_B respectively. The winner is determined by the distribution of units and B's type. For a given battle, A prevails with probability $p(\theta_i, \alpha', \beta')$, where i is B's type, and α' and β' give the current distribution of units. The only assumption that we make is that $p(\theta_S, \alpha', \beta') \leq p(\theta_W, \alpha', \beta')$ for all α' and β' , with the inequality holding strictly for some values. That is, A is less weakly likely to prevail in a given battle against the stronger opponent than the weaker opponent, and there is some military distribution where the stronger type is assuredly favored. One unit is eliminated from the loser's holdings, and a new round of bargaining begins. This process continues until B accepts a proposal or one side runs out of units and concedes the good.

The following proposition tells us when separating equilibria are not possible:

Proposition 4. *Suppose for a given distribution of units α' and β' that the battlefield*

results for the current period are not a function of B's type. Then the types do not separate in that period in equilibrium.

Essentially, Proposition 1's result and general intuition extends to a richer setting. If there is no differential risk in fighting the current period's battle, then A cannot successfully induce the weak type to separate from the strong type.

The basic idea of the proof is as follows. If the types will separate, then the strong type must reject and the weak type must accept. In the following period, A settles with the strong type—it knows which type it is facing, and any further fighting creates unnecessary inefficiency. However, this means that the weak type can obtain its aspirational payoff if it deviates to fighting. The conditions of Proposition 4 tell us that the weak type cannot resist that temptation when the outcome of the upcoming battle is not a function of its private information. If waiting a period is profitable for the strong type, then it must also be profitable for the weak type.

There are a couple of other important points in comparing Proposition 1 to Proposition 4. First, the absence of meaningful learning in the two-period model is not due to the single opportunity to acquire information before the climactic battle. If there are n battles before the first opportunity to learn, the types will never separate through the duration. Second, unlike the two-period case, Proposition 4 does not foreclose on the possibility of clean separation halfway through the interaction. For example, if a pivotal period exists after some set number of battles but well before the sides exhaust their troops, separating may be possible. This was not the case in the two-period model by construction, as no middling period existed there.

4.2 Uncertainty over Costs

In the main model, the actors are uncertain about the distribution of power in the second battle. Here we develop an identical model but switch the uncertainty to B's cost.

Explicitly, Nature draws B as a high cost type with probability $q \in (0, 1)$ and as a low cost type with probability $1 - q$. Similar to the main model, bargaining occurs over two stages, with actors proceeding to the second stage only if A wins the first battle. The only difference is that for both stages, the distribution of power remains the same, while A is uncertain about B's cost of fighting in the second stage. Therefore, if B

rejects in either stage, the actors fight a battle where A wins with probability $p \in [0, 1]$. In the first stage battle, A pays a cost $c_A > 0$ and B pays $c_B > 0$; the latter is common knowledge. In the second stage battle, A continues to pay c_A , whereas the low cost type of B pays $\underline{c}_B > 0$ and the high cost type pays $\bar{c}_B > \underline{c}_B$. Thus, like before, the source of uncertainty only manifests itself in the second stage.

This extension produces qualitatively identical results to the main model. No separating equilibria exist. Instead, A's strategy depends on its prior belief. When q is low, A guarantees B's acceptance in the first stage, analogous to Proposition 2; when q is high, A makes risky demands in both stages, analogous to Proposition 3. Once more, if war occurs with positive probability in the first stage, then it must also occur with positive probability in the second stage.

4.3 Flipping When Uncertainty Matters

What might happen if the uncertainty in the two battles is flipped? Sometimes, actors may know how the *second* phase would unfold but remain uncertain about their capacity to win the first phase. Unlike the main model, A does not receive meaningful new information, given that the second stage has complete information. Therefore, it is worthwhile to assess how the equilibrium and the probability of war might change if we flip the timing of the uncertainty.

Formally, in the first period, A now prevails against the weak type with probability $p'_1 \in (0, 1]$, while it prevails against the strong type with probability $p_1 \in [0, p'_1)$; A wins the second stage battle with the publicly-known probability $p_2 \in [0, 1]$.

A key difference arises as we solve the interaction. Because the types are not payoff relevant in the second stage, we can use backward induction to solve it. The standard result from the ultimatum game arises: A offers just enough to induce B to accept. Specifically, the amount demanded is $x_2 = p_2 + c_B$. Consequently, *no* war occurs in the second stage. This makes sense: the source of the bargaining friction is incomplete information, but the point at which that uncertainty matters has already passed.

Moving to the initial stage, the game effectively becomes another ultimatum game, though this time with more complicated payoffs and incomplete information. Nevertheless, the basic risk-return logic still governs A's proposal strategy. The appendix demonstrates that A prefers the risky demand if:

$$q > q^* \equiv \frac{c_A + c_B}{c_A + c_B + (p'_1 - p_1)(p_2 + c_B)} \quad (5)$$

Note that there are no complicated signaling dynamics here—B has nothing meaningful to communicate because its private information is irrelevant to the later stage. We therefore have a one-period war against just the low cost type if Line 5 holds and no war otherwise.

With that out of the way, we can compare the outcomes of the two games. Between the two models, there are six distinct probabilities of an actor winning a battle. However, there is an intuitive comparison. For uncertain battles, let $\bar{p} = p'_1 = p'_2$ represent the probability that A prevails against the weak type and $\underline{p} = p_1 = p_2$ represent the probability that A prevails against the strong type. In other words, we make the uncertain probabilities uniform across the models. We also define p as A's probability of winning in a battle with complete information so that is uniform as well.

After making those substitutions, the cutpoint of the previous model can be transformed as follows:

$$q_1^* \equiv \left(\frac{c_A + c_B}{c_A + c_B + \bar{p} - \underline{p}} \right) \left(\frac{c_A + c_B + (1 + p)(\bar{p} - \underline{p})}{c_A + c_B + p(\bar{p} - \underline{p})} \right)$$

We also have the cutpoint for the flipped model as follows:

$$q_2^* \equiv \frac{c_A + c_B}{c_A + c_B + (\bar{p} - \underline{p})(p + c_B)}$$

It is easy to prove that $q_1^* > q_2^*$; that is, the cutpoint of the original model for any war to occur is more stringent.⁴ Figure 4 displays the number of battles in the flipped model and the main model. Compared to a scenario where uncertainty over power exists only in the first battle, uncertainty in the second battle makes actors less likely to engage in war. However, once war breaks out, uncertainty in the second battle leads to more intense fighting. Since the war equilibrium for the original model includes the probabilities of fighting in both battles, and it is impossible for the flipped model to see a second battle, the original model is more likely to result in a war involving two battles than the flipped one.

⁴For proof, $q_1^* \equiv \left(\frac{c_A + c_B}{c_A + c_B + \bar{p} - \underline{p}} \right) \left(\frac{c_A + c_B + (1 + p)(\bar{p} - \underline{p})}{c_A + c_B + p(\bar{p} - \underline{p})} \right) > \frac{c_A + c_B}{c_A + c_B + (\bar{p} - \underline{p})p} > \frac{c_A + c_B}{c_A + c_B + (\bar{p} - \underline{p})(p + c_B)} \equiv q_2^*$

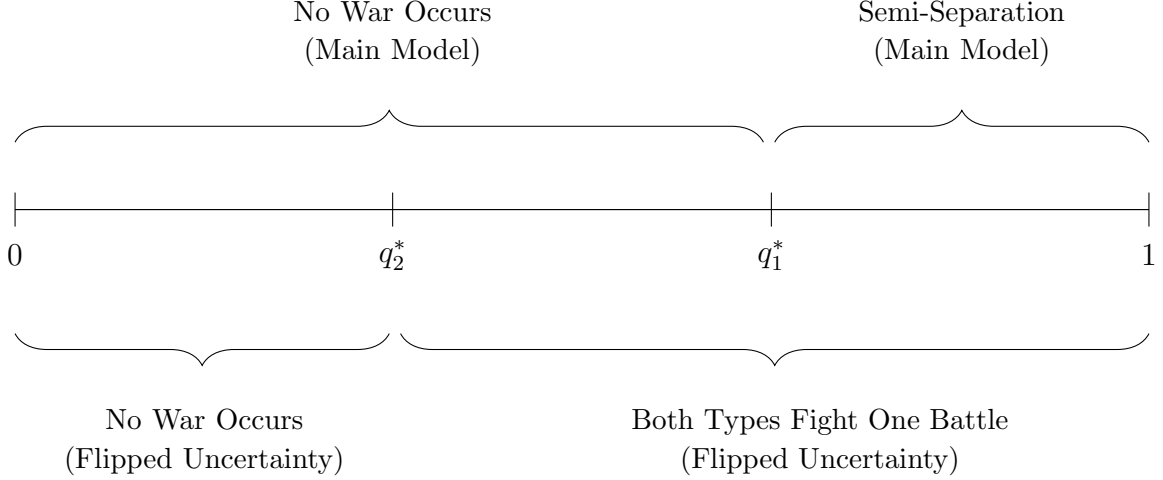


Figure 4: Prior Belief and the Number of Battles (Main Model & Flipped Uncertainty)

Intuitively, in the main model, semi-separation from the weak type is expensive. Only a portion will accept the first demand, and the strong type will reject all the way through. Consequently, A must have a strong belief that B is weak for it to go to that trouble. In contrast, A knows that any war in the flipped model will only last one period. It can try to separate the types at a low cost. If the strategy fails, then it can extract the surplus out of both types afterward. Consequently, the main model is more likely to produce an efficient outcome. However, when war does occur in it, the inefficiency is worse.

4.4 Semi-Informative First Battles

The main model set the possible probabilities of victory in the first battle as identical. A natural question to ask is whether the absence of a separating equilibrium only occurs on that boundary. To answer that, the appendix explores an extension where the probability of victory in the first battle is p_1 against the strong type and p'_1 against the weak type, where $0 < p_1 < p'_1 < 1$. This nests our main model when $p_1 = p'_1$ and nests the standard assumption of convergence models when $p_1 = p_2$.

We show that if p_1 is close enough to p'_1 , then no separating equilibrium exists for any value of q . In other words, Proposition 1 does not rely on a knife edge assumption that $p_1 = p'_1$. Intuitively, when they get close, separation requires paying the weak type

a quantity similar to the strong type while still inducing the strong type to fight in the first stage. Simply buying out both types in the first stage therefore becomes the most attractive option when q is low, and pursuing the semi-separating strategy is the most attractive option when q is high. The middle ground depicted in Figure 1 fails to exist.

The main takeaway here is that the first stage need not be identical for meaningful learning to fail. Rather, the first-stage outcomes just need to be similar enough. For example, if a type with the ability to handle long-distance logistics also has small advantages over short distances, that will still fail to produce a separating equilibrium.

5 Case Study: The Shining Path

Thus far, we have explored formal results that link latter-period uncertainty to prolonged wars. To further ground these ideas, we now transition to a case study of the Shining Path, or *Sendero Luminoso* in Spanish, and its battle against the Peruvian government in the 1980s and 1990s. Our goal here is to create an “existence proof” of the mechanism. To do this, we map the parameters of the game onto a substantive case and examine whether the thinking of the actors matches what the equilibrium predicts (Goemans and Spaniel 2016). The main point is that the Shining Path and the Peruvian government could not come to terms due to the type of uncertainty we have explored. Specifically, the Peruvian government plays the informed role of actor B from our model, having better knowledge of how the war would unfold once it reached urban centers. Meanwhile, the Shining Path plays the uninformed role of actor A from our model. Ultimately, the latter encounters the trouble that our semi-separating equilibrium highlights, where it is unable to meaningfully update its information given the government’s willingness to fight over the periphery.

Here, we do this in four phases. We begin with an overview of the conflict between the Shining Path and Peru to situate the case within existing narratives. Second, we substantively motivate the informational structure of the case—that an asymmetry existed about the latter stages of the conflict that the earlier stages of the conflict could not answer. The purpose here is to place the case within the scope conditions. We then trace how our model’s mechanism makes sense of the outcome reached in the war—that is, to map the established scope conditions to the equilibrium. Finally, we explore how our model better captures some of the dynamics of the conflict than the

standard commitment problem narrative does.

5.1 Overview

The Shining Path emerged as a significant guerrilla group in Peru that initiated one of the most violent conflicts in the country’s history. Founded by Abimael Guzmán, a former university professor, the group was heavily influenced by Maoist ideology. It aimed to overthrow the Peruvian government and establish a communist regime. The Shining Path began its insurgency in the late 1960s but escalated its operations significantly in 1980, coinciding with Peru’s transition back to democratic elections after a period of military rule.⁵

Initially, the Shining Path concentrated its efforts in the rural regions of Ayacucho and the neighboring departments of Apurímac and Huancavelica—areas largely neglected by Peru’s central government. This neglect provided fertile ground for recruitment and expansion. During President Fernando Belaúnde’s administration (1980–1985), the government’s response was relatively restrained, focusing more on containment than on aggressive countermeasures. Misjudging the threat, Belaúnde relied primarily on the National Police for counterinsurgency efforts instead of the army, a strategy that proved inadequate. Meanwhile, the Shining Path intensified its campaign by targeting local authorities and infrastructure, aiming to dismantle the state’s presence in the region. The group became notorious for its brutal tactics, including public executions and bombings. As the conflict escalated, the Shining Path expanded into other rural areas, gradually building a broader base of operations and support.

During Alan García’s presidency (1985–1990), there was an initial attempt to reform counterinsurgency practices by adopting a more integrated strategy that encompassed military, socio-political, and economic elements (Taylor 2006). This new approach focused on rural development and aimed to address the underlying causes of the insurgency. Efforts were also made to reform the intelligence services to improve coordination and effectiveness against the Shining Path. However, the effectiveness of García’s policies was severely constrained by the economic crises of the late 1980s, marked by hyperinflation and economic downturn. These financial challenges undermined state

⁵Though Shining Path used terrorism as a part of its tactics Kydd and Walter (2006); Spaniel (2019), it is coded as a civil war by major datasets Gleditsch et al. (2002), and we focus on that aspect of the conflict here.

capacity, exacerbating public discontent and limiting the government's ability to implement effective counterinsurgency strategies, which in turn contributed to the continued strength of the Shining Path.

In the late 1980s and early 1990s, the conflict took a new turn as the Shining Path extended its activities into urban centers, including the capital city, Lima. This shift significantly escalated the conflict by bringing it to the national forefront and exposing the group's radical agenda to a broader segment of Peruvian society. Under President Alberto Fujimori (1990–2000), the Peruvian government escalated its counterinsurgency measures, adopting increasingly aggressive and often brutal tactics. This strategic shift included leveraging enhanced intelligence capabilities and enacting stringent anti-terrorism legislation. Fujimori's government also engaged in significant autocratic backsliding as a means to cripple the insurgency, which included suspending the legislature and imposing significant restrictions on the press's ability to cover the war (Cameron 1998; Conaghan 2002). This consolidation of power facilitated more coordinated and decisive actions against the Shining Path. A critical outcome of these intensified efforts was the capture of Guzmán in 1992, which marked a turning point in the conflict by leading to the dismantling of key insurgent structures and a significant decline in the group's activities.

The Shining Path's rise in the 1980s, with its absence of negotiations and continued warfare, is often attributed to its extreme Maoist ideology. The historical narrative paints them as uncompromising hardliners, committed to an all-or-nothing approach in dismantling all existing state structures and considering anyone not aligned with them as an adversary (Gorriti 2000; Taylor 2006). This suggests that their ideological fervor left no space for compromise or negotiation.⁶ However, this explanation is incomplete. While ideology played a role, it alone does not explain why there was an inefficient mechanism of conflict resolution. Hence, the Shining Path's ideological stance does not

⁶While there were no direct negotiations between the Shining Path and the Peruvian government during the 1980s and early 1990s, there was a bargaining process outside the negotiation table. After battlefield gains in rural areas, Shining Path leader Abimael Guzmán made a more aggressive claim at the organization's first congress in 1988, announcing that it needed to shift its pursuit from the stage of strategic defense—marked by sabotage, propaganda, and guerrilla warfare—to a phase of strategic balance, implying a move toward nationwide conventional warfare against the government (Comisión de la Verdad y Reconciliación 2013). The bargaining took the form of the Peruvian government making no concession to the Shining Path's claim shift and intensifying its counter-insurgency activities in the formerly controlled rural areas, which resulted in an escalation of the urban battles.

fully unravel the complexity behind the sustained conflict and the lack of negotiation attempts.

Moreover, the conflict between the Shining Path and the Peruvian government challenges another prevalent theory: that prolonged wars facilitate the accumulation of information, reduce uncertainty, and lead toward resolution. As such, we ought to further explore why the case unfolded as it did.

5.2 Uncertainty over Urban Battles

With the historical overview out of the way, we now begin exploring how the Shining Path's case follow the logic of the model. The preliminary step of this process is to examine the connection between the case and the model's scope conditions (Goemans and Spaniel 2016). For our model in particular, we are interested in segmented uncertainty over the likelihood of victory in earlier versus later portions of a potential war.

In the Shining Path insurgency in Peru, a key element of the conflict dynamics was the Peruvian government's possession of optimistic private beliefs regarding potential outcomes of *urban* battles, specifically in Lima. Hence the government mimics B from the model, whereas the uninformed Shining Path mimics A. The initial period of fighting along the periphery represents battles that were identical across types, as in the first period of the model. The later urban period of fighting represents the case where types mattered, as in the second period of the model.

To substantiate that, we begin by exploring the source of the government's optimistic prior beliefs. The basis was two-fold: an understanding of public sentiment and superior military capabilities in urban settings. First, consider the urban-rule divide in public opinion. The Shining Path garnered support in rural regions due to the state's inability to respond effectively to peasant needs and demands. In contrast, the urban populace, especially in Lima, exhibited a markedly lower level of disenchantment with the government. A 1991 survey, which is from the latter stages of the conflict and is the most frequently cited poll, revealed that only about 12 percent of respondents in Lima expressed support for the Shining Path.⁷ This figure contrasted sharply with rural regions, where support was significantly higher, especially in areas like Ayacucho, with

⁷This figure was calculated by combining a 7 percent direct positive response with an additional estimated 5 percent, which represents half of the 10 percent of participants who did not respond, as recommended by the director of the survey company (McClintock 1998).

estimates suggesting sympathy for the group potentially as high as 80 percent (McClintock 1998; Taylor 2006). Indeed, the rural populace, feeling neglected and sharing grievances, was more receptive to the Maoist ideology of the Shining Path.

The key assumption in the model is not just a difference between the battle spaces, but rather a source of private information regarding the second stage. Here, the government's proximity and access to its urban support centers inherently gave it a clearer picture about its ability to withstand a Shining Path assault on the city. Recognizing the disparity, the government strategically focused its efforts, understanding that the Shining Path would encounter less support in diverse, less isolated urban settings. However, without the same proximity and access, the Shining Path could not form as complete of a picture. After some time, the Shining Path believed they had made sufficient progress in the countryside and were ready to win the cities (Arce Borja and Talavera Sánchez 1988).

Furthermore, the government was confident about its enhanced military capabilities in urban areas, bolstered by two main factors. First, the urban setting offered better visibility and less scope for guerrilla tactics like hit-and-run assaults, which the Shining Path had effectively employed in rural areas. Second, the government's intelligence capabilities were significantly more effective in urban areas. Rural regions presented challenges like language barriers and difficult terrain that hindered intelligence operations. In contrast, these barriers were notably reduced in urban settings, enhancing the government's ability to gather intelligence and respond strategically.

The *ex post* data regarding the number of casualties in Lima also lends support to the idea that state security forces were more present and efficient in the center. In Lima, the state was responsible for approximately 2,000 deaths, compared to the Shining Path's 800, a reversal of the pattern observed in rural areas where the Shining Path had been more deadly, without any exception (Comisión de la Verdad y Reconciliación 2013). This suggests that, while the Shining Path was more lethal across other regions and the state in general, the state's security forces in Lima were not only more efficient but also more brutal, demonstrating their relative strength and ruthlessness in urban warfare. Again, though, this was only made possible by high support for the government in the city, which the Shining Path was not informed of.

Zooming out, this also suggests that the battles in the rural peripheries did not offer the Shining Path insights that were directly transferable to urban warfare. The success

of the Shining Path in these peripheral regions was largely due to tactics and dynamics specific to rural guerrilla warfare. They exploited the terrain, the local populace's grievances against the central government, and the minimal state presence in these areas. However, these elements were largely absent or markedly different in urban settings.

That is, the transition to urban warfare required a significant tactical shift—just as the transition from the first to the second period in the model. Moreover, it was one that the Shining Path was not prepared for. Urban combat involved more direct confrontations and visible acts of terrorism, which differed significantly from the guerrilla tactics employed in the rural areas. Additionally, the urban populace's alignment and support were not as straightforward as in the rural peripheries. The Shining Path had leveraged the government's oversight and local disenchantment in the rural areas, but in Lima, the situation was different—the state's presence was stronger, and public sentiment was not as easily swayed by the Shining Path's ideology.

This disparity between the rural and urban combat environments meant that the successes and strategies employed by the Shining Path in the rural areas did not provide a reliable framework for predicting their performance in urban conflicts. For example, the inability for state military capacity to extend to the periphery was consistent with both a localized and general inability. The government understood this disconnect, contributing to its strategic advantage in the urban phase of the conflict. Conversely, the Shining Path's inability to adapt to the distinct urban conditions and its failure to accurately assess the government's readiness and public support in urban areas ultimately contributed to its decline.

5.3 Why Bargaining Failure?

The prior section established how the central scope conditions of our model are present in the case. Equilibrium is a rigorous way to map scope conditions to empirical predictions. We now examine whether there is a match here *and* whether the reasoning follows the equilibrium logic. Thus, the main things we are looking for are a prolonged war due to the later-stage optimistic private information causing the uninformed party to be too belligerent.

The core of our paper's mechanism lies in the uncertainty inherent in the second

stage of conflict and the absence of separating equilibrium. The model elucidates that when there is a source of uncertainty about an adversary’s capabilities in a new phase of conflict, parties are unable to segregate based on their types. This uncertainty was stark in the case of the Peruvian government and the Shining Path, where the latter’s success in rural areas did not provide clear insights into the urban warfare that awaited them. The Peruvian government, optimistic about its military capabilities, especially in urban areas, refused to back down to the Shining Path. Lacking the same information, the Shining Path believed that continuing to press would be wise; in the language of the model, it had a high prior belief q .

Indeed, we can tie the government’s strategy to its optimism. Interviews and statements from state security forces during this period reflect this sentiment. In one such interview, a military official stated, “We knew our military force was *far superior*” (Yates 2005, emphasis added). In the language of the model, the government was drawn as a stronger type. Indeed, this strategic advantage of the government ultimately culminated in the capture of Guzmán in Lima. However, the challenge of operating in the periphery, as the official explained, laid in the difficulty of gathering intelligence in rural areas, where few spoke the local language, Quechua, and distinguishing between the Shining Path insurgents and civilians, as they often concealed themselves among the local population (Yates 2005). These constraints were largely absent in urban warfare. Therefore, this statement highlights the contrast in the government’s strengths between rural and urban areas and helps to elucidate their inclination towards a more aggressive and war-prone stance in the conflict.

In summary, the failure of and prolongation of the conflict between the Shining Path and the Peruvian government can be attributed to the persistent uncertainties and differential risks associated with the transition from rural to urban warfare. Our model elucidates how these factors, when combined with the government’s access to private information, can overshadow the potential benefits of negotiation, prompting aggressive demands and the continuation of hostilities in multiple stages in the long run, even in the face of escalating costs.

5.4 Was It a Commitment Problem?

One may ask whether the protracted conflict in Peru can be attributed to a commitment problem, which is a common explanation for prolonged civil wars. However, applying the commitment problem framework to the Shining Path insurgency in Peru does not fully capture the complexities of this specific conflict. Unlike scenarios where commitment problems are prominent, the geographic context of the Peruvian conflict provided a clear basis for potential regional autonomy. The Shining Path's operations were primarily situated in the mountainous regions, suggesting a feasible option for establishing control in these areas without necessitating expansion into central regions.

That is, the expansion of the Shining Path's activities into central areas, including the capital, Lima, was a consequence of optimism rather than a necessity for survival. This move indicates a shift from a defensive stance to an offensive strategy aimed at governmental overthrow. This is crucial for contrasting the Peruvian conflict with other civil wars. While conflicts in regions like Syria, Libya, or Somalia are muddled by indistinct geographic boundaries, Peru presented a different scenario. Similar to the clear demarcations in the US Civil War—which allowed for the possibility of credible commitments to peace but also did not transpire due to strategic decisions—Peru's geography initially suggested a potential for regional autonomy. However, the Shining Path's decision to push beyond their mountain strongholds into urban centers reflects a departure from a path that could have relied on geographic divisions to facilitate a peaceful resolution.

Therefore, while the inability to commit to peace agreements is a significant factor in many civil conflicts, the Shining Path's insurgency in Peru suggests that other dynamics were at play. The group's decision to make greater demands out of the government, particularly their aims to reach urban centers, reflect ambitions that extend beyond mere minimal territorial control or survival. This indicates that the prolongation of the conflict was driven more by the Shining Path's gamble under uncertainty rather than commitment problems typically highlighted in the literature.

6 Conclusion

Many wars occur over distinct phases, and the underlying martial ability to win during one part may not matter during another. This paper took that observation as a point of departure from existing models of bargaining while fighting. Unfortunately, we discovered a negative result: that it is more difficult to resolve conflicts because proposing states cannot devise bargaining strategies to reveal types. Thus, we have established an information-based explanation for prolonged conflicts.

Our paper speaks to a broader need for conflict scholars to engage with military strategists to develop more detailed models that produce new empirical predictions. At first pass, the idea of a war occurring over distinct phases does not sound like much of a departure from traditional models. However, formalizing the idea—and combining it with uncertainty—yielded fresh results. Examining other key aspects of military strategy may similarly prove fruitful.

References

- Arce Borja, Luis and Janet Talavera Sánchez. 1988. “La Entrevista Del Siglo: El Presidente Gonzalo Rompe El Silencio.” *El Diario* (496).
- Arena, Philip. 2013. “Costly signaling, resolve, and martial effectiveness.” Manuscript, University of Buffalo, SUNY.
- Cameron, Maxwell A. 1998. “Self-Coups: Peru, Guatemala, and Russia.” *Journal of Democracy* 9(1):125–139.
- Comisión de la Verdad y Reconciliación. 2013. “Anexo 2: ¿Cuántos Peruanos Murieron? Estimación del total de víctimas causadas por el conflicto armado interno entre 1980 y el 2000.”.
- Conaghan, Catherine M. 2002. “Cashing in on Authoritarianism: Media collusion in Fujimori’s Peru.” *Harvard International Journal of Press/Politics* 7(1):115–125.
- Fearon, James D. 2004. “Why do some civil wars last so much longer than others?” *Journal of Peace Research* 41(3):275–301.
- Filson, Darren and Suzanne Werner. 2002. “A bargaining model of war and peace: Anticipating the onset, duration, and outcome of war.” *American Journal of Political Science* 46(4):819–837.

- Gleditsch, Nils Petter, Peter Wallensteen, Mikael Eriksson, Margareta Sollenberg and Håvard Strand. 2002. "Armed conflict 1946-2001: A new dataset." *Journal of peace research* 39(5):615–637.
- Goemans, Hein and William Spaniel. 2016. "Multimethod research: A case for formal theory." *Security Studies* 25(1):25–33.
- Gorriti, Gustavo. 2000. *The Shining Path: A History of the Millenarian War in Peru*. University of North Carolina Press.
- Krainin, Colin, Caroline Thomas and Thomas Wiseman. 2020. "Rational quagmires: Attrition, learning, and war." *Quarterly Journal of Political Science* 15.
- Kydd, Andrew H and Barbara F Walter. 2006. "The strategies of terrorism." *International Security* 31(1):49–80.
- McClintock, Cynthia. 1998. *Revolutionary Movements in Latin America: El Salvador's FMLN and Peru's Shining Path*. US Institute of Peace Press.
- Paine, Jack and Scott A Tyson. 2020. "Uses and abuses of formal models in political science."
- Philpott, William. 2015. *War of Attrition: Fighting the First World War*. New York: Abrams.
- Powell, Robert. 2004. "Bargaining and Learning while Fighting." *American Journal of Political Science* 48(2):344–361.
- Powell, Robert. 2006. "War as a commitment problem." *International organization* 60(1):169–203.
- Slantchev, Branislav L. 2003. "The Principle of Convergence in Wartime Negotiations." *American Political Science Review* 97(4):621–632.
- Smith, Bradley C and William Spaniel. 2019. "Militarized disputes, uncertainty, and leader tenure." *Journal of Conflict Resolution* 63(5):1222–1252.
- Spaniel, William. 2019. "Rational overreaction to terrorism." *Journal of Conflict Resolution* 63(3):786–810.
- Spaniel, William and Peter Bils. 2018. "Slow to learn: Bargaining, uncertainty, and the calculus of conquest." *Journal of Conflict Resolution* 62(4):774–796.
- Taylor, Lewis. 2006. *Shining Path: Guerrilla War in Peru's Northern Highlands*. Liverpool University Press.
- Walter, Barbara F. 1997. "The Critical Barrier to Civil War Settlement." *International Organization* 51(3):335–364.

Yates, Pamela. 2005. "State of Fear: The Truth about Terrorism.". Documentary.

Appendix

We now prove all of the claims from the paper.

A.1 Proof of Main Model

We begin with some preliminaries about the second stage. Let $r^* \equiv \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2}$. Then the following lemma gives the A's stage two equilibrium actions:

Lemma 1. *If $r > r^*$, A proposes $x_2 = p'_2 + c_B$. The weak type accepts and the strong type rejects. If $r < r^*$, A proposes $x_2 = p_2 + c_B$. Both types accept. When $r = r^*$, A is indifferent between those proposals.*

Proof: Each type must accept when its payoff exceeds its war payoff. Given that A will win against the weak type with probability p'_2 and against the strong type with probability p_2 , the war payoff for the weak type is $1 - p'_2 - 2c_B$ and for the strong type is $1 - p_2 - 2c_B$.¹ To induce the weak type to accept in the second stage, we have:

$$\begin{aligned} 1 - x_2 - c_B &\geq 1 - p'_2 - 2c_B \\ x_2 &\leq p'_2 + c_B \end{aligned}$$

To make the strong type accept, we have:

$$\begin{aligned} 1 - x_2 - c_B &\geq 1 - p_2 - 2c_B \\ x_2 &\leq p_2 + c_B \end{aligned}$$

If A proposes $x_2 > p'_2 + c_B$, both types reject and A receive the war payoff. A will always deviate since the war payoff is its worst option. If A proposes a conservative demand $x_2 \leq p_2 + c_B$, both types accept.² To optimize its payoff, A will propose the highest feasible value $x_2 = p_2 + c_B$. If A proposes an aggressive demand $x_2 \in (p_2 + c_B, p'_2 + c_B]$, the strong type rejects and the weak type accepts, and A's payoff

¹After entering into the second stage, actors face a sunk cost due to the fighting in the first stage. As a result, if B fights a war in the second stage, it should pay the cost of war twice. Similarly, the accept payoff is $1 - x_2 - c_B$ due to the sunk cost.

²The strong type is indifferent between rejecting and accepting when $x_2 = p_2 + c_B$. Under this circumstances, it must accept in the equilibrium, otherwise there is a best response problem.

depends on its posterior belief r that B is the weak type: $rx_2 + (1-r)(p_2 - c_A) - c_A$. To optimize its payoff, A will propose the highest feasible value $x_2 = p'_2 + c_B$. To induce A to make an aggressive demand, we have:

$$r(p'_2 + c_B) + (1-r)(p_2 - c_A) - c_A > p_2 + c_B - c_A$$

$$r > \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \equiv r^*$$

Therefore, A will make the aggressive demand $x_2 = p'_2 + c_B$ when $r > r^*$, make the conservative demand $x_2 = p_2 + c_B$ when $r < r^*$, and is indifferent when $r = r^*$.

We now turn to the first stage. There are three phases to this. The first two are simple. We can begin by deriving x_1 proposals that guarantee that both types will accept. Next, we can show proposals that guarantee that both types will reject. The middle range of proposals depends on beliefs and updating and makes up the bulk of the work for the proof.

To begin, consider the guaranteed acceptance case:

Lemma 2. *Both types accept $x_1 \leq p_1(p_2 + c_B) + c_B$.*

The guaranteed rejection case is as follows:

Lemma 3. *Both types reject $x_1 > p_1(p'_2 + c_B) + c_B$.*

Proof: In the second stage, the strong type must receive its war payoff in any equilibrium. If A makes the conservative demand, the strong type is indifferent between peace and war. If A makes the aggressive one, the strong type rejects and starts the battle. Given that the B wins the first battle with probability $1 - p_1$ no matter its type, the strong type's payoff after rejection in the first stage is $1 - p_1(p_2 + c_B) - c_B$. To induce the strong type to accept in the first stage, we have:

$$1 - x_1 \geq 1 - p_1(p_2 + c_B) - c_B$$

$$x_1 \leq p_1(p_2 + c_B) + c_B$$

For the weak type, even though it always accepts whether A makes an aggressive or a conservative demand in the second stage, the payoff varies. If A makes an aggressive demand $x_2 = p'_2 + c_B$ in the second stage, the weak type receives a lower payoff of

rejection $1 - p_1(p'_2 + c_B) - c_B$ in the first stage. If A instead makes a conservative demand $x_2 = p_2 + c_B$ in the second stage, the weak type receives a higher payoff of rejection $1 - p_1(p_2 + c_B) - c_B$ in the first stage. Therefore, the weak type must accept in the first stage when:

$$1 - x_1 \geq 1 - p_1(p_2 + c_B) - c_B$$

$$x_1 \leq p_1(p_2 + c_B) + c_B$$

Conversely, the weak type must reject in the first stage when:

$$1 - x_1 < 1 - p_1(p'_2 + c_B) - c_B$$

$$x_1 > p_1(p'_2 + c_B) + c_B$$

Since $p_1(p'_2 + c_B) + c_B > p_1(p_2 + c_B) + c_B$, the strong type must also reject when $x_1 > p_1(p'_2 + c_B) + c_B$.

A.1.1 Proof of Proposition 1

For the middle range of proposals $x_1 \in (p_1(p_2 + c_B) + c_B, p_1(p'_2 + c_B) + c_B]$, the strong type always rejects and receives the war payoff $1 - p_1(p_2 + c_B) - c_B$. Suppose the separating equilibrium exists. Then the weak type must act differently and accept the offer.

If A survives the war and enters into the second stage, it will have its posterior belief r equal 0 and consider B as a strong type. As a result, A will make a conservative demand $x_2 = p_2 + c_B$.

However, the weak type would prefer to deviate to fighting and receive the higher war payoff $1 - p_1(p_2 + c_B) - c_B$. Therefore, there is no separating equilibrium for the main model.

A.1.2 Proof of Proposition 2 and 3

First, there is no pooling equilibrium when the proposals are in the middle range $x_1 \in (p_1(p_2 + c_B) + c_B, p_1(p'_2 + c_B) + c_B]$. For proof, the strong type always rejects when $x_1 > p_1(p_2 + c_B) + c_B$ and a pooling equilibrium exists only when the weak type follows

the strong type and rejects the offer. In this situation, the weak type receives the war payoff.

If $q > r^*$, A has its posterior belief $r = q > r^*$ and makes an aggressive demand $x_2 = p'_2 + c_B$ in the second stage. The weak type receives $1 - p_1(p'_2 + c_B) - c_B$ after rejecting the first stage proposal. In this situation, the weak type has a profitable deviation. Had it instead accepted the offer, it would always get a payoff $1 - x_1$. This is better if:

$$1 - x_1 > 1 - p_1(p'_2 + c_B) - c_B$$

$$x_1 < p_1(p'_2 + c_B) + c_B$$

If $q < r^*$, A has its posterior belief $r = q < r^*$ and makes a conservative demand $x_2 = p_2 + c_B$ in the second stage. The weak type receives $1 - p_1(p_2 + c_B) - c_B$ after rejecting the first stage proposal. In this situation, the weak type has no profitable deviations since $1 - x_1 \leq 1 - p_1(p_2 + c_B) - c_B$. However, A will always deviate to make the conservative demand $x_1 = p_1(p_2 + c_B) + c_B$ in the first stage. For proof, A receives $p_1(p_2 + c_B) - c_A$ if both types reject in the first stage and A makes the conservative demand in the second stage. Then we always have:

$$p_1(p_2 + c_B) - c_A < p_1(p_2 + c_B) + c_B$$

Second, based on Lemma 2, both types accept when the proposal $x_1 \leq p_1(p_2 + c_B) + c_B$. In this situation, A receives its payoff under agreement x_1 . To maximize its payoff, A demands $x_1 = p_1(p_2 + c_B) + c_B$. This conservative first-stage demand leads to the equilibrium of Proposition 2, in which no war occurs.

Nevertheless, A will deviate to make a more aggressive demand when it holds a strong belief that B is a weak type. There are two types of deviations A could make: First, A makes an extremely aggressive proposal $x_1 > p_1(p'_2 + c_B) + c_B$, including a pooling strategy where both types reject. Given that B is highly likely to be the weak type, the war payoff for A is still greater than what it can get from the agreement. Second, A makes a proposal in the middle range $x_1 \in (p_1(p_2 + c_B) + c_B, p_1(p'_2 + c_B) + c_B]$, including a semi-separating strategy where the strong type rejects but the weak type mixes between accepting and rejecting.

We begin with the case in which war always occurs. If A proposes $x_1 > p_1(p'_2 +$

$c_B) + c_B$, according to Lemma 3, both types reject. In this situation, A receives the war payoff based on its belief of the probability B is a weak type. Given that both types reject in the first stage, if A enters into the second stage, its posterior belief r equals the prior belief q . If $q < r^*$, from Lemma 1 we know that A proposes $x_2 = p_2 + c_B$ in the second stage and both types accept the offer. Then A will not deviate from the conservative proposal given the war payoff $p_1(p_2 + c_B) - c_A$ is less than the payoff under agreement $p_1(p_2 + c_B) + c_B$. If $q > r^*$, however, A proposes $x_2 = p'_2 + c_B$ in the second stage. This time, A receives $p'_2 + c_B$ in the second stage when B is more likely to be the weak type and accepts the offer, and $p_2 - c_A$ when B is more likely to be the strong type and rejects the offer. A's payoff when fighting a war with both types in the first stage becomes $p_1[q(p'_2 + c_B) + (1 - q)(p_2 - c_A)] - c_A$. Let $q^{**} \equiv \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \right) \left(\frac{1 + p_1}{p_1} \right)$. Then A prefers to fighting both sides in the first stage when $q > q^{**}$:

$$p_1[q(p'_2 + c_B) + (1 - q)(p_2 - c_A)] - c_A > p_1(p_2 + c_B) + c_B$$

$$q > \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \right) \left(\frac{1 + p_1}{p_1} \right)$$

In a moment, we will prove that the payoff for inducing semi-separation is better when q is larger than the equilibrium cutpoint q^* . Moreover, q^{**} must be greater than $q^* = \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \right) \left(\frac{c_A + c_B + (1 + p_1)(p'_2 - p_2)}{c_A + c_B + p_1(p'_2 - p_2)} \right)$, thus covering this case.³

Now we consider the second situation, in which the proposal lies in the middle range. If A proposes $x_1 \in (p_1(p_2 + c_B) + c_B, p_1(p'_2 + c_B) + c_B]$, even though there is neither pooling equilibrium nor separating equilibrium. It is still possible that B have a semi-separating strategy with the strong type rejecting and the weak type mixing between accepting and rejecting.

Given that the strong type always rejects in the first stage, the semi-separating equilibrium exists only when the weak type mixes between accepting and rejecting in the first stage, and A is indifferent between making the conservative demand and the aggressive demand in the second stage. Let σ_R be the probability the weak type rejects in the first stage, and $1 - \sigma_R$ be the probability the weak type accepts. Based on the Bayes' Rule, we can have A's posterior belief that B is the weak type as a function of q and σ_R : $\frac{q\sigma_R}{q\sigma_R + (1 - q)}$. From Lemma 1, A is indifferent between the two proposals in the

³ $q^{**} > q^* > r^*$. For proof, let $k^* \equiv \frac{c_A + c_B}{p'_2 - p_2}$. q^* can be transformed to $q^* = (1 + \frac{1}{k^* + p_1})r^*$ and q^{**} can be transformed to $q^{**} = (1 + \frac{1}{p_1})r^*$. Since $\frac{1}{p_1} > \frac{1}{k^* + p_1} > 0$, $q^{**} > q^* > r^*$.

second stage if $r = r^*$. Solving for σ_R :

$$\frac{q\sigma_R}{q\sigma_R + (1-q)} = \frac{c_A + c_B}{p'_2 - p_2 + c_A + c_B}$$

$$\sigma_R = \left(\frac{1-q}{q} \right) \left(\frac{c_A + c_B}{p'_2 - p_2} \right) \equiv \sigma_R^*$$

This is 0-1 constrained for the parameter space in which the weak type plays it.⁴

Second, let $\sigma_R^* \equiv \left(\frac{1-q}{q} \right) \left(\frac{c_A + c_B}{p'_2 - p_2} \right)$. We have A's payoff in the first stage as follows:

$$q(1 - \sigma_R^*)x_1 + [1 - q(1 - \sigma_R^*)][p_1(p_2 + c_B) - c_A]$$

To optimize its payoff, A will set x_1 to its largest feasible value $p_1(p'_2 + c_B) + c_B$. Ultimately, A's payoff becomes:

$$q(1 - \sigma_R^*)[p_1(p'_2 + c_B) + c_B] + [1 - q(1 - \sigma_R^*)][p_1(p_2 + c_B) - c_A]$$

Finally, the semi-separating equilibrium exists only when A has no profitable deviation. To make A prefer the aggressive demand $x_1 = p_1(p'_2 + c_B) + c_B$ to the conservative one $x_1 = p_1(p_2 + c_B) + c_B$, we have:

$$q(1 - \sigma_R^*)[p_1(p'_2 + c_B) + c_B] + [1 - q(1 - \sigma_R^*)][p_1(p_2 + c_B) - c_A] > p_1(p_2 + c_B) + c_B$$

$$q > \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \right) \left(\frac{c_A + c_B + (1 + p_1)(p'_2 - p_2)}{c_A + c_B + p_1(p'_2 - p_2)} \right) \equiv q^*$$

To make A prefer the aggressive demand $x_1 = p_1(p'_2 + c_B) + c_B$ to the war demand $x_1 > p_1(p'_2 + c_B) + c_B$, we have:

$$q(1 - \sigma_R^*)[p_1(p'_2 + c_B) + c_B] + [1 - q(1 - \sigma_R^*)][p_1(p_2 + c_B) - c_A] > p_1[q(p'_2 + c_B) + (1 - q)(p_2 - c_A)] - c_A$$

⁴Note that $\sigma_R^* \in (0, 1]$ when $q \geq r^*$. For proof, σ_R^* is always larger than 0. To make σ_R^* less than or equal to 1, we have:

$$\left(\frac{1-q}{q} \right) \left(\frac{c_A + c_B}{p'_2 - p_2} \right) \leq 1$$

$$q \geq \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2}$$

$$q > \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \equiv r^*$$

Since the cutpoint of the prior belief q that makes the pooling equilibrium of both types rejecting exist is $q^{**} > q^* > r^*$, the semi-separating strategy following the aggressive demand $x_1 = p_1(p'_2 + c_B) + c_B$ always gives A a greater payoff than the extremely aggressive demand $x_1 > p_1(p'_2 + c_B) + c_B$.

Therefore, when A's prior belief that B is the weak type is sufficiently high, it makes an aggressive demand $x_1 = p_1(p'_2 + c_B) + c_B$ in the first stage. The strong type rejects and the weak type rejects with a probability $\sigma_R^* \equiv \left(\frac{1-q}{q}\right) \left(\frac{c_A+c_B}{p'_2-p_2}\right)$. In the second stage, A demands $x_2 = p'_2 + c_B$. The strong type rejects and the weak type accepts. The cutpoint that makes A prefers the aggressive demand $x_1 = p_1(p'_2 + c_B) + c_B$ to the conservative demand $x_1 = p_1(p_2 + c_B) + c_B$ is:

$$q > \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2}\right) \left(\frac{c_A + c_B + (1 + p_1)(p'_2 - p_2)}{c_A + c_B + p_1(p'_2 - p_2)}\right) \equiv q^*$$

Therefore, we have proved Proposition 3:

Proposition 3 (Full Version). *Suppose A's prior belief that B is weak is sufficiently high (i.e., $q > q^*$). Then A makes an aggressive first-stage demand, $x_1 = p_1(p'_2 + c_B) + c_B$. The strong type always rejects, while the weak type rejects with probability $\sigma_R^* \equiv \left(\frac{1-q}{q}\right) \left(\frac{c_A+c_B}{p'_2-p_2}\right)$. In the second stage, A then demands an amount that only the weak type accepts, $x_2 = p'_2 + c_B$, and the strong type fights a second battle.*

The two off-the-equilibrium-path first-stage demands are as follows.

1) *If A instead makes a conservative first-stage demand, $x_1 = p_1(p_2 + c_B) + c_B$, both types pool by accepting and no war occurs.*

2) *If A instead makes a war demand in the first stage, $x_1 > p_1(p'_2 + c_B) + c_B$, both types pool by rejecting. In the second stage, A demands an amount that only the weak type accepts, $x_2 = p'_2 + c_B$, and the strong type fights a second battle.*

Meanwhile, A prefers the opposite when the inequality is flipped, as the propositions claimed (Proposition 2):

Proposition 2 (Full Version). *Suppose A's prior belief that B is weak is sufficiently low (i.e., $q < q^*$). Then A makes a conservative first-stage demand, $x_1 = p_1(p_2 + c_B) + c_B$. Both types pool by accepting and no war occurs.*

Though neither type enters Stage 2 on the equilibrium path, the relevant off-equilibrium-path contingencies are as follows: (a) when q is moderately low (i.e., $r^* < q < q^*$), A demands an amount that only the weak type accepts, $x_2 = p'_2 + c_B$, and the strong type rejects and fights a second battle; (b) when q is extremely low (i.e., $q < r^*$), A makes a conservative demand, $x_2 = p_2 + c_B$, such that both types accept and no war occurs.

The two off-the-equilibrium-path first-stage demands are as follows.

1) If A instead makes an aggressive first-stage demand, $x_1 = p_1(p'_2 + c_B) + c_B$, The strong type always rejects, while the weak type mixes between accepting and rejecting. In the second stage, A then demands an amount that only the weak type accepts, $x_2 = p'_2 + c_B$, and the strong type fights a second battle.

2) If A instead makes a war demand in the first stage, $x_1 > p_1(p'_2 + c_B) + c_B$, both types pool by rejecting. In the second stage, A demands an amount that only the weak type accepts, $x_2 = p'_2 + c_B$, and the strong type fights a second battle.

Table A.1 shows the cutpoints, demand sizes, posterior beliefs, and bargaining outcomes for both equilibria.

Equilibrium	Cutpoint	S_1 Demand	S_2 Demand	Posterior Belief	S_1 Outcome	S_2 Outcome
Proposition 2	$q < q^*$	$p_1(p_2 + c_B) + c_B$	Off-the-Path: i) $p'_2 + c_B$ ($r^* < q < q^*$); and ii) $p_2 + c_B$ ($q < r^*$).	Off-the-Path: q	No war occurs	Off-the-Path: i) Strong type fights; weak type accepts ($r^* < q < q^*$); and ii) No war occurs ($q < r^*$).
Proposition 3	$q > q^*$	$p_1(p'_2 + c_B) + c_B$	$p'_2 + c_B$	$\frac{c_A + c_B}{p'_2 - p_2 + c_A + c_B}$	War occurs with probability $(1 - q)(1 + \frac{c_A + c_B}{p'_2 - p_2})$; Strong type fights; weak type fights with probability σ_R^*	Strong type fights; weak type accepts

Table A.1: Summary of Equilibria

A.2 Proof of Proposition 4

First, the strong type must receive its war payoff in any equilibrium. For proof, suppose an equilibrium exists where at some accept/reject decision it has, its payoff strictly exceeds its war payoff. Consider the last of those sequentially within the extensive form. The only way the strong type can beat its war payoff at that node is through a proposal that exceeds that value. Because this is the final node that results in such a

proposal, the strong type earns its war payoff if it rejects. Therefore, it must accept the proposal. The weak type earns weakly less through war and therefore must also accept. In turn, A's payoff is the quantity it demanded.

However, A has a profitable deviation. Consider a proposal that still gives the strong type more than its war payoff but strictly less than the hypothesized equilibrium proposal—e.g., the midpoint between those two values. Both types must still accept, but now A keeps slightly more for itself. As such, there can be no final node on the path where the strong type earns more than its war payoff. That, in turn, guarantees that the strong type earns its war payoff starting at any node on the path.

Second, in any separating equilibrium, a deal must be immediately reached in the following stage. This is because a separating equilibrium means that A knows through common knowledge of strategies the type of B that rejected. Let θ be that type, $V_A(\theta)$ be A's continuation value for no agreement at that point given that type, and $V_B(\theta)$ be the revealed type's continuation value. Because both players suffer costs for this period's battle and the sum of payoffs maximize at 1, $V_A(\theta) + V_B(\theta) < 1$. Then, in the period following separation, the revealed type accepts if $1 - x_t \geq V_B(\theta)$, or $x_t \leq 1 - V_B(\theta)$, where t denotes the period. A's optimal acceptable demand is therefore $x_t = 1 - V_B(\theta)$. This is better than inducing rejection if $1 - V_B(\theta) > V_A(\theta)$, or $V_A(\theta) + V_B(\theta) < 1$, which is true.⁵ Thus, in the period following separation, the separating type earns $V_B(\theta)$.

Third, in any separating equilibrium, the weak type must earn its war payoff in the next period if it is the type that rejects. The proof is analogous to the first step. At the last of the nodes where the weak type earns more than its war payoff, A could deviate to demanding more for itself and still induce the weak type to accept.

Fourth, in a separating equilibrium, the strong type must reject and the weak type must accept. Suppose the reverse is true. Then the weak type earns its war payoff at the separating period—it is fighting at that period, and the third part showed that the weak type earns its war payoff afterward. Meanwhile, the strong type must earn at least its war payoff by accepting. If the types have different war payoffs at this stage, then the weak type could profitably deviate to accepting. If the types have identical war payoffs at this stage, then A's proposal must give B an amount equal to that identical war payoff to induce both types to reject. However, by analogous logic to the first part

⁵No equilibria exist where the type rejects with positive probability because of the standard best response problem from the ultimatum game.

of the proof, the equilibrium proposals for the rest of the game must continue to induce indifference from B. In turn, both types must accept anything that pays strictly greater than its war payoff. By making a slightly larger concession, A can therefore induce both types to accept with certainty. This is better than suffering its own inefficiency c_A and not extracting B's inefficiency c_B with probability q . Consequently, the only separating equilibrium can involve the strong type rejecting and the weak type accepting.

Finally, we can derive a profitable deviation for A. Suppose entering the period, A has α' units and B has β' units. To induce the weak type to accept in period t , we have:

$$1 - x_t \geq p(\theta_W, \alpha', \beta')w(\theta_S, \alpha', \beta' - 1) + (1 - p(\theta_W, \alpha', \beta'))w(\theta_S, \alpha' - 1, \beta') - c_B$$

$$x_t \leq 1 - p(\theta_W, \alpha', \beta')w(\theta_S, \alpha', \beta' - 1) - (1 - p(\theta_W, \alpha', \beta'))w(\theta_S, \alpha' - 1, \beta') + c_B$$

And to induce the strong type to reject, we have:

$$1 - x_t \leq p(\theta_S, \alpha', \beta')w(\theta_S, \alpha', \beta' - 1) + (1 - p(\theta_S, \alpha', \beta'))w(\theta_S, \alpha' - 1, \beta') - c_B$$

$$x_t \geq 1 - p(\theta_S, \alpha', \beta')w(\theta_S, \alpha', \beta' - 1) - (1 - p(\theta_S, \alpha', \beta'))w(\theta_S, \alpha' - 1, \beta') + c_B$$

If there is no differential in the battle outcome probabilities at α' and β' , then $p(\theta_W, \alpha', \beta') = p(\theta_S, \alpha', \beta')$. As such, the proposal must be:

$$x_t = 1 - p(\theta_S, \alpha', \beta')w(\theta_S, \alpha', \beta' - 1) - (1 - p(\theta_S, \alpha', \beta'))w(\theta_S, \alpha' - 1, \beta') + c_B$$

That is, the amount that makes the strong type indifferent between accepting and rejecting. With probability q , the weak type accepts, and A earns that amount instantly. With probability $1 - q$, the strong type rejects, and A extracts all of the surplus out of the strong type in the next period after suffering its cost in the current period.

However, A has a profitable deviation analogous to the fourth stage. If it proposes any more than the hypothesized equilibrium quantity, both types must accept. (This gives the strong type a settlement larger than its war payoff, which cannot happen in

any further stage. The weak type accepts as well because it cannot do better than the strong type's war payoff.) The slightly larger concession still generates more for A because it is not suffering its own inefficiency c_A and is extracting some portion of B's costs c_B with probability q .

A.3 Proof of Uncertainty over Costs

A.3.1 Equilibrium

Similar to the proof of the main model, we begin with the preliminaries about the second stage. A's demand in the second stage depends on its posterior belief about B's type. If A holds a strong belief that B is a high cost type and will accept the offer, it will take the risk and make an aggressive demand in the second stage. If A instead holds a weak posterior belief, it will make a concession to ensure the low cost type's compliance. Letting r be A's posterior belief that B is the high cost type in the second stage, A will make a risky demand if:

$$r > \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B} \quad (6)$$

Proof: Each type must accept when its payoff exceeds that of war. In the second stage, the low cost type pays $\underline{c}_B$ to fight the second battle, resulting in a war payoff of $1 - p - \underline{c}_B$. The war payoff for the high cost type is $1 - p - \bar{c}_B$. To induce the high cost type to accept in the second stage, we have:

$$1 - x_2 - c_B \leq 1 - p - \underline{c}_B - c_B$$

$$x_2 \leq p + \underline{c}_B$$

To make the low cost type accept, we have:

$$1 - x_2 - c_B \leq 1 - p - \bar{c}_B - c_B$$

$$x_2 \leq p + \bar{c}_B$$

If A proposes $x_2 > p + \bar{c}_B$, both types reject and A receives the war payoff. A will always deviate since the war payoff is its worst option. If A proposes a conservative

demand $x_2 \leq p + \underline{c}_B$, both types accept.⁶ A will then propose its highest feasible value $x_2 = p + \underline{c}_B$ to optimize the payoff. If A instead proposes an aggressive demand $x_2 \in (p + \underline{c}_B, p + \bar{c}_B]$, the low cost type rejects and the high cost type accepts. In this situation, A's payoff depends on its posterior belief r that B is the high cost type: $r(x_2) + (1 - r)(p - c_A) - c_A$. Since this payoff increases with the level of demand, A will propose the highest feasible value $x_2 = p + \bar{c}_B$. To induce A to make an aggressive demand, we have:

$$r(p + \bar{c}_B) + (1 - r)(p - c_A) - c_A > p + \underline{c}_B - c_A$$

$$r > \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B}$$

Let $r_{uc}^* \equiv \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B}$. A will make the aggressive demand $x_2 = p + \bar{c}_B$ when $r > r_{uc}^*$, make the conservative demand $x_2 = p + \underline{c}_B$ when $r < r_{uc}^*$, and is indifferent when $r = r_{uc}^*$. We then have the lemma for A's stage two equilibrium actions:

Lemma 4. *If $r > r_{uc}^*$, A proposes $x_2 = p + \bar{c}_B$. The weak type accepts and the strong type rejects. If $r < r_{uc}^*$, A proposes $x_2 = p + \underline{c}_B$. Both types accept. When $r = r_{uc}^*$, A is indifferent between those proposals.*

Returning to the first stage, similar to the main model with uncertainty over power, A faces a decision between a safety demand and an aggressive demand. Analogous to the main model, if A has a considerably low expectation about B being a high cost type, it will make a conservative proposal, both types accept. If A instead has an extremely high expectation about B being a high cost type, it will demand a proposal extremely high in the first stage, both types reject. Therefore, we can derive the x_1 proposals that guarantee that both types will accept and reject.

Lemma 5. *Both types accept $x_1 \leq p(p + \underline{c}_B) + c_B$.*

Lemma 6. *Both types reject $x_1 > p(p + \bar{c}_B) + c_B$.*

Proof: Just as the strong type in the main model, the low cost type must receive its war payoff in the second stage. If A makes the conservative demand, the low cost type

⁶The low cost type is indifferent between rejecting and accepting when $x_2 = p + \underline{c}_B$. Under this circumstances, it must accept in the equilibrium, otherwise there is a best response problem.

is indifferent between peace and war. If A makes the aggressive one, the low cost type rejects and starts the battle. Given that the first battle costs c_B regardless of type, B's stage one war payoff is $1 - p(p + \underline{c}_B) - c_B$. To induce the low cost type to accept in the first stage, we have:

$$1 - x_1 \geq 1 - p(p + \underline{c}_B) - c_B$$

$$x_1 \leq p(p + \underline{c}_B) + c_B$$

For the high cost type, even though it always accepts whether A makes an aggressive or a conservative demand in the second stage, the payoff varies. If A makes an aggressive demand $x_2 = p + \bar{c}_B$ in the second stage, the high cost type receives a lower stage one war payoff $1 - p(p + \bar{c}_B) - c_B$. If A instead makes a conservative demand $x_2 = p + \underline{c}_B$ in the second stage, the high cost type receives a higher stage one war payoff $1 - p(p + \underline{c}_B) - c_B$. Therefore, the high cost type must accept in the first stage when:

$$1 - x_1 \geq 1 - p(p + \underline{c}_B) - c_B$$

$$x_1 \leq p(p + \underline{c}_B) + c_B$$

Conversely, the high cost type must reject in the first stage when:

$$1 - x_1 < 1 - p(p + \bar{c}_B) - c_B$$

$$x_1 > p(p + \bar{c}_B) + c_B$$

Since $p(p + \bar{c}_B) + c_B > p(p + \underline{c}_B) + c_B$, the low cost type must follow the high cost type and reject when A makes an extremely aggressive demand $x_1 > p(p + \bar{c}_B) + c_B$.

In the middle range: $x_1 \in (p(p + \underline{c}_B) + c_B, p(p + \bar{c}_B) + c_B]$, the implications are qualitatively identical to the main model: There is no separating equilibrium exists.

Proposition 5. *No separating equilibrium exists.*

Also, the only existing equilibrium in the middle range is the semi-separating equilibrium, where A holds a relatively high expectation of B being a high cost type and consequently demands a comparatively aggressive proposal in the first stage.

We can derive our equilibria similar to the main model. Only two parameter spaces exist with

$$q_{uc}^* \equiv \max\left\{ \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B}, \left(\frac{c_A + c_B}{c_A + \bar{c}_B} \right) \left(\frac{c_A + c_B + (1+p)(\bar{c}_B - \underline{c}_B)}{c_A + c_B + p(\bar{c}_B - \underline{c}_B)} \right) \right\} \quad (7)$$

serving as the cutpoint that lies between them.

Proposition 6. *Suppose A's prior belief that B faces high battle cost is sufficiently low. Then A makes a conservative demand in the first stage. Both types accept.*

Proposition 7. *Suppose A's prior belief that B faces high battle cost is sufficiently high. Then A makes an aggressive demand in the first stage. The low cost type always rejects, while the high cost type mixes between accepting and rejecting. In the second stage, A makes another aggressive demand. The high cost type always accepts, while the low cost type always rejects.*

Figure A.1 illustrates the two parameter spaces of the model as a function of q and c_A . Similar to the main model, when q is small and c_A is large, A is sufficiently convinced that B is the low cost type and A's cost of execute a noisy screening of types is large. As a result, A will make a conservative proposal to induce both types to accept and the Proposition 6's logic wins out.

When q is large and c_A is small, A is sufficiently convinced that B is the high cost type and A's cost of battle is small. As a result, A will not make large concessions at the start. Instead, A is willing to induce some inefficiencies in the first stage and will make an aggressive proposal that holds the high cost type to its war payoff. The low cost type always rejects and the high cost type sometimes rejects in the first stage. Proposition 7's logic wins out.

A.3.2 Proof of Proposition 5

For the middle range: $x_1 \in (p(p + \underline{c}_B) + c_B, p(p + \bar{c}_B) + c_B]$, the low cost type always rejects and receives the war payoff $1 - p(p + \underline{c}_B) - c_B$. Suppose the separating equilibrium exists. Then the high cost type must act differently and accept the offer. A will demand $x_1 = p(p + \bar{c}_B) + c_B$ to optimize its payoff, and the weak type receives $1 - p(p + \bar{c}_B) - c_B$.

If A survives the war and enters into the second stage, it will have its posterior belief r equal 0 and consider B as a low cost type. As a result, A will make a conservative demand $x_2 = p + \underline{c}_B$.

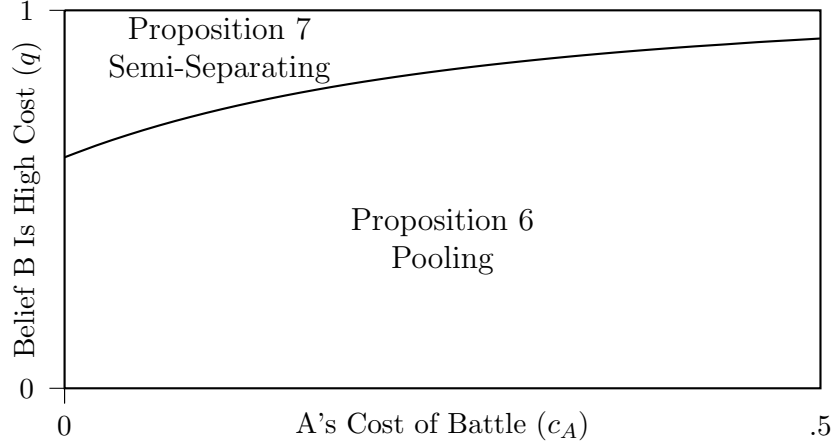


Figure A.1: Equilibrium plot of the model with uncertainty over cost. The line demarcating the two parameter spaces is q_{uc}^* , as defined on Line 7.

However, the high cost type will always deviate to fighting in the first stage and receive the higher payoff for a second stage settlement $1 - p(p + \underline{c}_B) - c_B$. Therefore, there is no separating equilibrium.

A.3.3 Proof of Proposition 6

First, there is no pooling equilibrium when the proposals are in the middle range $x_1 \in (p(p + \underline{c}_B) + c_B, p(p + \bar{c}_B) + c_B]$. For proof, the low cost type always rejects when $x_1 > p(p + \underline{c}_B) + c_B$. In a pooling equilibrium, the high cost type follows the low cost type and rejects the offer. In this situation, the high cost type receives its war payoff.

If $q > r_{uc}^*$, A has its posterior belief $r = q > r_{uc}^*$ and makes an aggressive demand $x_2 = p + \bar{c}_B$ in the second stage. The high cost type receives $1 - p(p + \bar{c}_B) - c_B$ after rejecting the first stage proposal. In this situation, the high cost type has a profitable deviation. Had it instead accepted the offer, it would always get a payoff $1 - x_1$, which is better than its war payoff. We can easily prove it from the range of x_1 :

$$\begin{aligned} x_1 &< p(p + \bar{c}_B) + c_B \\ 1 - x_1 &> 1 - p(p + \bar{c}_B) - c_B \end{aligned}$$

If $q < r_{uc}^*$, A has its posterior belief $r = q < r_{uc}^*$ and makes a conservative demand

$x_2 = p + \underline{c}_B$ in the second stage. The high cost type receives $1 - p(p + \underline{c}_B) - c_B$ after rejecting the first stage proposal. In this situation, the high cost type has no profitable deviations since $1 - x_1 \leq 1 - p(p + \underline{c}_B) - c_B$. However, A will always deviate to make the conservative demand $x_1 = p(p + \underline{c}_B) + c_B$ in the first stage. For proof, A receives $p(p + \underline{c}_B) - c_A$ if both types reject in the first stage and A makes the conservative demand in the second stage. Then we have:

$$p(p + \underline{c}_B) - c_A < p(p + \underline{c}_B) + c_B$$

Second, based on Lemma 5, both types accept proposals $x_1 \leq p(p + \underline{c}_B) + c_B$. In this situation, A receives its payoff under agreement x_1 . To optimize the utility, A demands $x_1 = p(p + \underline{c}_B) + c_B$.

Nevertheless, A will deviate to make a more aggressive demand when it holds a strong belief that B is a high cost type. There are two situations that A may deviate. First, A makes an aggressive proposal $x_1 > p(p + \bar{c}_B) + c_B$ and there is a pooling strategy where both types reject. Given that B is sufficiently likely to be the high cost type, the war payoff for A is still greater than what it can get from the agreement. Second, A makes a proposal in the middle range $x_1 \in (p(p + \underline{c}_B) + c_B, p(p + \bar{c}_B) + c_B]$, and there is a semi-separating strategy that the low cost type rejects but the high cost type mixes between accepting and rejecting.

If A proposes $x_1 > p(p + \bar{c}_B) + c_B$, according to Lemma 6, both types reject. In this situation, A receives the war payoff based on its belief of the probability B is a high cost type. Given that both types reject in the first stage, if A enters into the second stage, its posterior belief r equals the prior belief q . If $q \leq r_{uc}^*$, from Lemma 4 we know that A proposes $x_2 = p + \underline{c}_B$ in the second stage and both types accept the offer. Then A will not deviate from the conservative proposal given the war payoff $p(p + \underline{c}_B) - c_A$ is less than the payoff under agreement $p(p + \underline{c}_B) + c_B$. If $q > r_{uc}^*$, however, A proposes $x_2 = p + \bar{c}_B$ in the second stage. This time, A receives $p + \bar{c}_B$ in the second stage when B is the high cost type and accepts the offer and $p - c_A$ when B is the low cost type and rejects the offer. A's payoff when fighting a war with both types in the first stage becomes $p(p + q\bar{c}_B - (1 - q)c_A) - c_A$. Let $q_{uc}^{**} \equiv \frac{c_A + c_B + p(c_A + \underline{c}_B)}{p(c_A + \bar{c}_B)}$, A prefers to fighting both sides in the first stage when $q > q_{uc}^{**}$:

$$p(p + q\bar{c}_B - (1 - q)c_A) - c_A > p(p + \underline{c}_B) + c_B$$

$$q > \frac{c_A + c_B + p(c_A + \underline{c}_B)}{p(c_A + \bar{c}_B)}$$

If A proposes $x_1 \in (p(p + \underline{c}_B) + c_B, p(p + \bar{c}_B) + c_B]$, even though there is neither pooling equilibrium nor separating equilibrium. It is still possible that B have a semi-separating strategy with the low cost type rejecting and the high cost type mixing between accepting and rejecting.

Let $q_{uc}^* \equiv \left(\frac{c_A + c_B}{c_A + \bar{c}_B} \right) \left(\frac{c_A + c_B + (1 + p)(\bar{c}_B - \underline{c}_B)}{c_A + c_B + p(\bar{c}_B - \underline{c}_B)} \right)$. Here we also set a condition $c_B \leq \underline{c}_B$ for the comparison between different cutpoints of q , since the relationship remains uncertain between the first-stage war cost c_B and the second-stage war costs for the two types, $\underline{c}_B$ and $\bar{c}_B$. Under this condition, $q_{uc}^{**} > \max\{q_{uc}^*, r_{uc}^*\}$.⁷

We will prove in the next section that the semi-separating equilibrium exists when $q > \max\{q_{uc}^*, r_{uc}^*\}$.

Therefore, if A's prior belief that B is the high cost type is sufficient small, it will make a conservative demand $x_1 = p(p + \underline{c}_B) + c_B$. Both types accept.

A.3.4 Proof of Proposition 7

First, when the proposals are in the middle range $x_1 \in (p(p + \underline{c}_B) + c_B, p(p + \bar{c}_B) + c_B]$, the low cost type always reject in the first stage. In a semi-separating equilibrium, the high cost type mixes between accepting and rejecting in the first stage, and A is indifferent between making the conservative demand and the aggressive demand in the second stage. Let σ_R be the probability the high cost type reject in the first stage, and $1 - \sigma_R$ be the probability the high cost type accept. By Bayes' Rule, we can have A's posterior belief that B is the weak type as a function of q and σ_R : $\frac{q\sigma_R}{q\sigma_R + (1 - q)}$. From Lemma 4, A is indifferent between the two proposals in the second stage if $r = r_{uc}^*$. Solving for σ_R :

$$\frac{q\sigma_R}{q\sigma_R + (1 - q)} = \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B}$$

⁷The reason for setting this condition is to simplify the evaluation of the case to demonstrate that there is no separation in this model extension. In alternative cases where this condition does not hold, there is still no separating equilibrium, but the explicit solution is significantly more complex and is therefore omitted here.

$$\sigma_R = \left(\frac{1-q}{q} \right) \left(\frac{c_A + c_B}{\bar{c}_B - \underline{c}_B} \right)$$

Second, let $\sigma_R^{**} \equiv \left(\frac{1-q}{1} \right) \left(\frac{c_A + c_B}{\bar{c}_B - \underline{c}_B} \right)$.⁸ Then we have A's payoff in the first stage as follows:

$$q(1 - \sigma_R^*)x_1 + [1 - q(1 - \sigma_R^*)][p(p + \underline{c}_B) - c_A]$$

To optimize its payoff, A will set x_1 to its largest feasible value $p(p + \bar{c}_B) + c_B$. Ultimately, A's payoff becomes:

$$q(1 - \sigma_R^*)[p(p + \bar{c}_B) + c_B] + [1 - q(1 - \sigma_R^*)][p(p + \underline{c}_B) - c_A]$$

Finally, the semi-separating equilibrium exists only when A has no profitable deviation. To make A prefer the aggressive demand $x_1 = p(p + \bar{c}_B) + c_B$ to the conservative one $x_1 = p(p + \underline{c}_B) + c_B$, we have:

$$q(1 - \sigma_R^*)[p(p + \bar{c}_B) + c_B] + [1 - q(1 - \sigma_R^*)][p(p + \underline{c}_B) - c_A] > p(p + \underline{c}_B) + c_B$$

$$q > \left(\frac{c_A + c_B}{c_A + \bar{c}_B} \right) \left(\frac{c_A + c_B + (1+p)(\bar{c}_B - \underline{c}_B)}{c_A + c_B + p(\bar{c}_B - \underline{c}_B)} \right) \equiv q_{uc}^*$$

To make A prefer the aggressive demand $x_1 = p(p + \bar{c}_B) + c_B$ to the war demand $x_1 > p(p + \bar{c}_B) + c_B$, we have:

$$q(1 - \sigma_R^*)[p(p + \bar{c}_B) + c_B] + [1 - q(1 - \sigma_R^*)][p(p + \underline{c}_B) - c_A] > p(p + q\bar{c}_B - (1-q)c_A) - c_A$$

$$q > \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B} \equiv r_{uc}^*$$

Since the cutpoint of the prior belief q that makes the pooling equilibrium of both types rejecting exist is $q_{uc}^{**} > r_{uc}^*$, as long as the semi-separating strategy following the aggressive demand $x_1 = p(p + \bar{c}_B) + c_B$ exists, A will never demand $x_1 > p(p + \bar{c}_B) + c_B$ to induce both types to reject. However, unlike the main model, we cannot directly make

⁸ $\sigma_R^{**} \in (0, 1]$ when $q \geq r_{uc}^*$. For proof, σ_R^{**} is always larger than 0. To make σ_R^{**} less than or equal to 1, we have $q \geq \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B}$.

a comparison between r_{uc}^* and q_{uc}^* given that we are uncertain about the relationship between the stage one cost of war for B and that in the stage two. Therefore, we use the maximum value $\max\{q_{uc}^*, r_{uc}^*\}$ as the cutpoint for the semi-separating equilibrium. Under our previous condition for the power distribution $c_B \leq \underline{c}_B$, $q_{uc}^{**} > \max\{q_{uc}^*, r_{uc}^*\}$. As a result, we exclude the strategy that induces pooling on rejection because it generates a lower payoff than the strategy that induces semi-separation for any value of q .

Therefore, when A's prior belief that B is the high cost type is sufficiently high, it makes an aggressive demand $x_1 = p(p + \bar{c}_B) + c_B$ in the first stage. The low cost type rejects and the high cost type rejects with a probability $\sigma_R^{**} \equiv \left(\frac{1-q}{q}\right) \left(\frac{c_A + c_B}{\bar{c}_B - \underline{c}_B}\right)$. In the second stage, A demands $x_2 = p + \bar{c}_B$. The low cost type rejects and the high cost type accepts. The cutpoint that makes A prefer the aggressive demand $x_1 = p(p + \bar{c}_B)$ to the conservative demand $x_1 = p(p + \underline{c}_B)$ is:

$$q > \max\left\{ \frac{c_A + \underline{c}_B}{c_A + \bar{c}_B}, \left(\frac{c_A + c_B}{c_A + \bar{c}_B}\right) \left(\frac{c_A + c_B + (1+p)(\bar{c}_B - \underline{c}_B)}{c_A + c_B + p(\bar{c}_B - \underline{c}_B)}\right) \right\}$$

A.4 Proof of Flipped Uncertainty

We begin with situations in the second stage. Based on the model setting, both actors have complete information about the second battle. A wins with probability p_2 . The war payoffs for both types of B is $1 - p_2 - 2c_B$ and for A is $p_2 - 2c_A$. To induce B to accept in the second stage, we have:

$$1 - x_2 - c_B \geq 1 - p_2 - 2c_B$$

$$x_2 \leq p_2 + c_B$$

To optimize its payoff, A will propose the highest feasible value $x_2 = p_2 + c_B$. Since $p_2 + c_B - c_A > p_2 - 2c_A$, A will never deviate to propose a higher value to induce war. As a result, both types accept and no war happens in the second stage.

Back in the first stage, the strong types wins the first battle with probability $1 - p_1$, and the weak type wins with probability $1 - p_1'$. Since both types receive equal payoffs $1 - p_2 - 2c_B$ in the second stage, the strong type receives $1 - p_1(p_2 + c_B) - c_B$ after rejection in the first stage, and the weak type receives $1 - p_1'(p_2 + c_B) - c_B$. To induce

the strong type to accept in the first stage, we have:

$$1 - x_1 \geq 1 - p_1(p_2 + c_B) - c_B$$

$$x_1 \leq p_1(p_2 + c_B) + c_B$$

Analogously, the weak type accepts if:

$$1 - x_1 \geq 1 - p'_1(p_2 + c_B) - c_B$$

$$x \leq p'_1(p_2 + c_B) + c_B$$

If A proposes $x_1 > p'_1(p_2 + c_B) + c_B$, both type reject and A receives the war payoff $p_1(p_2 + c_B) - c_A$. A will always deviate to propose $x_1 \leq p'_1(p_2 + c_B) + c_B$ since the war is its worst option. If A proposes a conservative demand $x_1 \leq p_1(p_2 + c_B) + c_B$, both type accept. To optimize its payoff, A will propose the highest feasible value $x_1 = p_1(p_2 + c_B) + c_B$. If A proposes a middle-range demand $x_1 \in (p_1(p_2 + c_B) + c_B, p'_1(p_2 + c_B) + c_B]$, the strong type rejects and the weak type accepts, and A's payoff depends on its prior belief q : $qx_1 + (1 - q)(p_2 + c_B - c_A)$. To optimize its payoff, A will propose $x_1 = p'_1(p_2 + c_B) + c_B$. To induce A to take the risk, we have:

$$q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p_1(p_2 + c_B) - c_A] > p_1(p_2 + c_B) + c_B$$

$$q > \frac{c_A + c_B}{c_A + c_B + (p'_1 - p_1)(p_2 + c_B)}$$

We can derive the equilibrium as follows. As is shown in line 5, there are two parameter spaces exist with $q^* \equiv \frac{c_A + c_B}{c_A + c_B + (p'_1 - p_1)(p_2 + c_B)}$ serving as the cutpoint that lies between them. A will make the aggressive demand $x_1 = p'_1(p_2 + c_B) + c_B$ when $q > q^*$, make the conservative demand $x_1 = p_1(p_2 + c_B) + c_B$ when $q < q^*$, and become indifferent when $q = q^*$.

Proposition 8. *If line 5 holds, then A's belief that B is weak is sufficiently low. As a result, A make will a conservative demand in the first stage. Both types accept.*

Proposition 9. *If line 5 doesn't hold, then A's belief that B is weak is sufficiently high. As a result, A will make an aggressive demand in the first stage. The strong type always rejects, while the weak type always accepts. In the second stage, A makes its demand and both types accepts.*

A.5 Proof of the Semi-Informative First Battles

In the main model, the actors are uncertain about the distribution of power only for fighting the second battle, while having complete information about the first fighting. Here we develop a new model that considers the situation that the actors are uncertain about their power also for the both stages. To be explicit, if B rejects in the first stage, the actors fight the first battle. If Nature drew the weak type, A prevails with probability $p'_1 \in (0, 1]$; if Nature drew the strong type, A prevails with probability $p_1 \in [0, p'_1)$. The set-up for the second stage is identical to that in the main model.

A.5.1 Equilibrium

Similar to the proof of the main model, we begin with the preliminaries about the second stage. Given that this new model is identical to the main model in the second stage, the cutpoint for A's posterior belief r to induce A to make a risky demand is exactly equal to that in the main model:

$$r > \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2}$$

Returning to the first stage, analogous to the main model, there is a guaranteed acceptance case and a guaranteed rejection case:

Lemma 7. *Both types accept $x_1 \leq p_1(p_2 + c_B) + c_B$.*

Lemma 8. *Both types reject $x_1 > p'_1(p'_2 + c_B) + c_B$.*

Proof: Similar to the main model, the strong type must receive its war payoff in the second stage. Given that the strong type wins the first battle with probability $1 - p_1$, its payoff after rejection is $1 - p_1(p_2 + c_B) - c_B$. To induce the strong type to accept in the first stage, we have:

$$1 - x_1 \geq 1 - p_1(p_2 + c_B) - c_B$$

$$x_1 \leq p_1(p_2 + c_B) + c_B$$

For the weak type, the payoff depends on A's posterior belief. If A holds a sufficiently low posterior belief that B is the weak type, it makes an aggressive demand $x_2 = p'_2 + c_B$

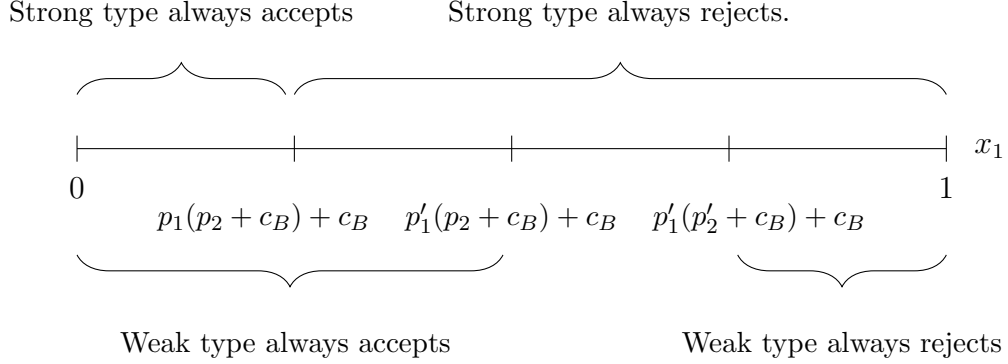


Figure A.2: First Stage Demands and Both Types' Responses

in the second stage. The weak type receives a lower payoff of rejection $1 - p'_1(p'_2 + c_B) - c_B$. If A instead holds a sufficiently high posterior belief and makes a conservative demand $x_2 = p_2 + c_B$ in the second stage, the weak type receives a higher payoff after rejection $1 - p'_1(p_2 + c_B) - c_B$. Therefore, the weak type must accept in the first stage when:

$$1 - x_1 \geq 1 - p'_1(p_2 + c_B) - c_B$$

$$x_1 \leq p'_1(p_2 + c_B) + c_B$$

Conversely, the weak type must reject in the first stage when:

$$1 - x_1 < 1 - p'_1(p'_2 + c_B) - c_B$$

$$x_1 > p'_1(p'_2 + c_B) + c_B$$

Since $p'_1(p'_2 + c_B) + c_B > p_1(p_2 + c_B) + c_B$, the weak type must follow the strong type and reject when $x_1 > p'_1(p'_2 + c_B) + c_B$. Figure A.2 illustrate both types' response conditional upon the first stage proposal x_1 .

For the middle range $x_1 \in (p_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$, the equilibrium is dependent upon the relationship between p'_1 and p_1 . If $\frac{p'_1}{p_1} < \left(\frac{p'_1 - p_1}{p'_2 - p_2} \right) (p_2 + c_B) \left[1 + \frac{(c_A + c_B)}{p'_1(p'_2 - p_2)} \right] \equiv R^*$, the implications are different from the main model. The separating equilibrium exists when A's prior belief q is at its middle level, which induces A to make the moderately aggressive demand $x_1 = p'_1(p_2 + c_B) + c_B$.

Proposition 10. *Suppose A's prior belief that B is the weak type is at a middle level.*

Then A makes a moderately aggressive demand $x_1 = p'_1(p_2 + c_B) + c_B$ in the first stage. The strong type accepts while the weak type rejects. In the second stage, A makes the conservative demand and the strong type accepts.

On the other hand, the semi-separating equilibrium exists when A 's prior belief q is sufficiently high, which induces A to make the aggressive demand $x_1 = p'_1(p'_2 + c_B) + c_B$.

Proposition 11. *Suppose A 's prior belief that B faces is the weak type is sufficiently high. Then A makes an aggressive demand $x_1 = p'_1(p'_2 + c_B) + c_B$ in the first stage. The strong type always rejects, while the weak type mixes between accepting and rejecting. In the second stage, A is indifferent between the aggressive and the conservative demand. The weak type always accepts, while the strong type sometimes fight a second battle.*

Similar to the main model, the pooling equilibrium occurs only when A 's prior belief q is sufficiently low, which induces A to compromise and guarantee that both types accept.

Proposition 12. *Suppose A 's prior belief that B is the weak type is sufficiently low. Then A makes a conservative demand $x_1 = p_1(p_2 + c_B) + c_B$ in the first stage. Both types accept.*

The cutpoint for q that induces A to shift from its conservative demand $x_1 = p_1(p_2 + c_B) + c_B$ to a moderately aggressive demand $x_1 = p'_1(p_2 + c_B) + c_B$ is equal to the cutpoint for the model with flipped uncertainty:

$$q > q_{U1}^* \equiv \frac{c_A + c_B}{c_A + c_B + (p_2 + c_B)(p'_1 - p_1)} \quad (8)$$

To make A shift from its moderately aggressive demand $x_1 = p'_1(p_2 + c_B) + c_B$ to an aggressive demand $x_1 = p'_1(p'_2 + c_B) + c_B$, the cutpoint for q is:

$$q > q_{U3}^* \equiv \frac{c_A + c_B}{c_A + c_B + \frac{p'_1(p'_2 - p_2)}{p_1 + \frac{p_1}{p'_1}(\frac{c_A + c_B}{p'_2 - p_2})}} \quad (9)$$

If $\frac{p'_1}{p_1} > \left(\frac{p'_1 - p_1}{p'_2 - p_2}\right)(p_2 + c_B) \left[1 + \frac{(c_A + c_B)}{p'_1(p'_2 - p_2)}\right]$, the separating equilibrium doesn't exist. Similar to the main model, the only existing equilibrium in the middle range $x_1 \in$

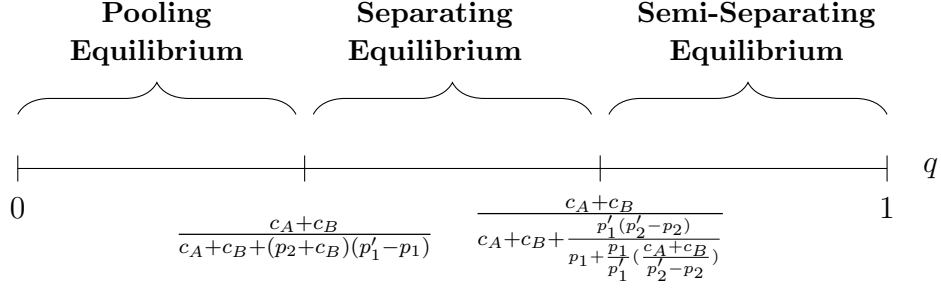


Figure A.3: Prior Belief and the Equilibrium ($\frac{p'_1}{p_1} < R^*$)

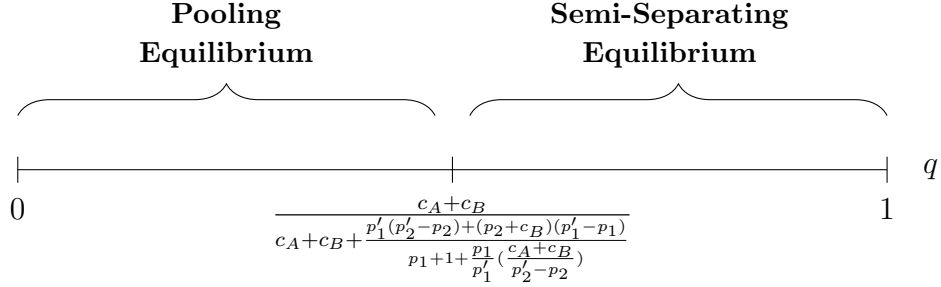


Figure A.4: Prior Belief and the Equilibrium ($\frac{p'_1}{p_1} > R^*$)

$(p_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$ is the semi-separating equilibrium. This threshold always holds in our main model.⁹

This time, the cutpoint for q that induces A to shift from its conservative demand $x_1 = p_1(p_2 + c_B) - c_B$ to its extremely aggressive demand $x_1 = p'_1(p'_2 + c_B) - c_B$ is:

$$q > q_{U2}^* \equiv \frac{c_A + c_B}{c_A + c_B + \frac{p'_1(p'_2 - p_2) + (p_2 + c_B)(p'_1 - p_1)}{p_1 + 1 + \frac{p_1}{p'_1} \left(\frac{c_A + c_B}{p'_2 - p_2} \right)}} \quad (10)$$

Figure A.3 represents the cutpoint for q when the separating equilibrium exists ($\frac{p'_1}{p_1} < R^*$), and Figure A.4 represents the situation that there is no separating equilibrium ($\frac{p'_1}{p_1} > R^*$). Notably, A does not need complete information for the separating equilibrium to fail to exist. As long as p_1 is close enough to p'_1 , the difference between the two types' first-stage power becomes small enough to make $\frac{p'_1}{p_1}$ converge to

⁹When there is complete information in the first stage, i.e., $p'_1 = p_1$, the left-hand side equals 1, which is greater than the right-hand side, which equals 0.

1 and R^* converge to 0. As a result, $\frac{p'_1}{p_1} > R^*$ always holds and the separating equilibrium no longer exist. Once $p_1 = p'_1$, the cutpoint for q between the pooling equilibrium and the semi-separating equilibrium, q_{U2}^* , is identical to that in the main model: $q^* \equiv \left(\frac{c_A + c_B}{c_A + c_B + p'_2 - p_2} \right) \left(\frac{c_A + c_B + (1 + p_1)(p'_2 - p_2)}{c_A + c_B + p_1(p'_2 - p_2)} \right)$.

A.5.2 Proof of Proposition 10

When $x_1 \in (p_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$, the strong type always rejects and receives the war payoff $1 - p_1(p_2 + c_B) - c_B$. Suppose the separating equilibrium exists, the weak type must act differently and accept the offer.

First, it is easy to prove that there is no separating equilibrium for the moderately high range: $x_1 \in (p'_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$. If the weak type deviates by rejecting the offer in the first stage, A will update its posterior belief to $r = 0$ that B is not the weak type, and will then make a conservative demand $x_2 = p_2 + c_B$. As a result, the weak type receives a payoff of $1 - p'_1(p_2 + c_B) - c_B$ by fighting the first war to secure a better bargaining outcome in the second stage. Since the weak type would have received $1 - x_1 < 1 - p'_1(p_2 + c_B) - c_B$ if it had accepted the first-stage proposal, it always has an incentive to deviate.

Second, for the moderately low range: $x_1 \in (p_1(p_2 + c_B) + c_B, p'_1(p_2 + c_B) + c_B]$, the separation occurs under some certain conditions of q . A demand $x_1 = p'_1(p_2 + c_B) + c_B$ to optimize its payoff, and the weak type receives $1 - p'_1(p_2 + c_B) - c_B$.

If A survives the war and enters into the second stage, it considers B as a strong type and makes a conservative demand $x_2 = p_2 + c_B$. The weak type never deviates. For proof, if the weak type fights in the first stage, it receives its war payoff with a better second stage settlement $1 - p'_1(p_2 + c_B) - c_B$, which is exactly identical for its payoff when accepting in the first stage. Since $1 - x_1 \geq 1 - p'_1(p_2 + c_B) - c_B$, we apply the refinement convention to ensure that the weak type, being indifferent or worse off from deviating, will never adopt this strategy.

Once the weak type adopts the separating strategy, A receives $q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p_1(p_2 + c_B) - c_A]$. To make A shift from the most conservative proposal $x_1 = p_1(p_2 + c_B) + c_B$ to a riskier proposal $x_1 = p'_1(p_2 + c_B) + c_B$, we have:

$$q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p_1(p_2 + c_B) - c_A] \geq p_1(p_2 + c_B) + c_B$$

$$q \geq \frac{c_A + c_B}{c_A + c_B + (p_2 + c_B)(p'_1 - p_1)} \equiv q_{U1}^*$$

Also, it is crucial to ensure that A never deviates to make the extremely high demand $x_1 > p'_1(p'_2 + c_B) + c_B$ and enjoy its payoff by fighting a war against both types. Later, we will show that under the pooling strategy, A updates its belief about B's type through a learning process based on the battlefield outcome: $r = \frac{(q)p'_1}{(q)p'_1 + (1-q)p_1}$. Therefore, for A to make the conservative demand $x_2 = p_2 + c_B$ in the second stage, $q \leq \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, which ensures a posterior belief of $r \leq \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$. As a result, A's payoff under separating strategy always dominates that under pooling on rejecting strategy $q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p_1(p_2 + c_B) - c_A] > q[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A]$. If $q > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, A instead makes the aggressive stage two demand $x_2 = p'_2 + c_B$. To induce A not to deviate, we have:

$$q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p_1(p_2 + c_B) - c_A] > q[p'_1(p'_2 + c_B) - c_A] + (1 - q)[p_1(p_2 - c_A) - c_A]$$

$$q < \begin{cases} \frac{(c_A + c_B)}{\frac{p_1 - 1}{p_1}(c_A + c_B) + \frac{p'_1}{p_1}(p'_2 - p_2)}, & \text{if } \frac{p'_1}{p_1} > \left(\frac{1 - p_1}{p_1}\right) \left(\frac{c_A + c_B}{p'_2 - p_2}\right) \\ 1, & \text{if } \frac{p'_1}{p_1} \leq \left(\frac{1 - p_1}{p_1}\right) \left(\frac{c_A + c_B}{p'_2 - p_2}\right) \end{cases}$$

When $\frac{p'_1}{p_1} > \left(\frac{p'_1 - p_1}{p'_2 - p_2}\right) (p_2 + c_B) + \frac{1}{p_1} \left(\frac{c_A + c_B}{p'_2 - p_2}\right)$, $\frac{(c_A + c_B)}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)} < \frac{(c_A + c_B)}{\frac{p_1 - 1}{p_1}(c_A + c_B) + \frac{p'_1}{p_1}(p'_2 - p_2)} < q_{U1}^*$. Therefore, the separating strategy is always dominated by either the pooling-on-accepting strategy or the pooling-on-rejecting strategy. As a result, no separating equilibrium exists.¹⁰

When $\frac{p'_1}{p_1} \in \left(\left(\frac{1 - p_1}{p_1}\right) \left(\frac{c_A + c_B}{p'_2 - p_2}\right), \left(\frac{p'_1 - p_1}{p'_2 - p_2}\right) (p_2 + c_B) + \frac{1}{p_1} \left(\frac{c_A + c_B}{p'_2 - p_2}\right)\right)$, $\frac{(c_A + c_B)}{\frac{p_1 - 1}{p_1}(c_A + c_B) + \frac{p'_1}{p_1}(p'_2 - p_2)} > q_{U1}^*$. As a result, the range of q for the separating strategy to dominate the pooling strategies is:

$$q \in [q_{U1}^*, \frac{(c_A + c_B)}{\frac{p_1 - 1}{p_1}(c_A + c_B) + \frac{p'_1}{p_1}(p'_2 - p_2)})$$

¹⁰This is a stricter condition than the one we later prove under which no separating equilibrium exists, i.e., $\frac{p'_1}{p_1} > R^*$. Specifically, when $\frac{p'_1}{p_1} > R^*$ holds, it is easy to show that $R^* < \left(\frac{p'_1 - p_1}{p'_2 - p_2}\right) (p_2 + c_B) + \frac{1}{p_1} \left(\frac{c_A + c_B}{p'_2 - p_2}\right)$, which implies that $\frac{p'_1}{p_1} > \left(\frac{p'_1 - p_1}{p'_2 - p_2}\right) (p_2 + c_B) + \frac{1}{p_1} \left(\frac{c_A + c_B}{p'_2 - p_2}\right)$ is a stricter condition.

When $\frac{p'_1}{p_1} < \left(\frac{1-p_1}{p_1}\right) \left(\frac{c_A+c_B}{p'_2-p_2}\right)$, the separating strategy dominates the pooling strategies when:¹¹

$$q \in [q_{U1}^*, 1)$$

A.5.3 Proof of Proposition 12

First, there is no pooling equilibrium when the proposals are in the moderately low range $x_1 \in (p_1(p_2 + c_B) + c_B, p'_1(p_2 + c_B) + c_B]$. For proof, the strong type always rejects when $x_1 > p_1(p_2 + c_B) + c_B$, and the weak type always accepts when $x_1 < p'_1(p_2 + c_B) + c_B$.

Second, there is no pooling equilibrium when the proposals are in the moderately high range $x_1 \in (p'_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$. In a pooling equilibrium, the weak type follows the strong type and rejects the offer. In this situation, the weak type receives its war payoff.

To induce A to make an aggressive demand in the second stage, its posterior belief that B is the weak type must be sufficiently high: $r > \frac{c_A+c_B}{c_A+c_B+(p'_2-p_2)} \equiv r^*$. Given that A updates its belief based on the battlefield outcome in the first stage: $r = \frac{(q)p'_1}{(q)p'_1+(1-q)p_1}$, we obtain:

$$\begin{aligned} \frac{(q)p'_1}{(q)p'_1+(1-q)p_1} &> \frac{c_A+c_B}{c_A+c_B+(p'_2-p_2)} \\ q &> \frac{c_A+c_B}{c_A+c_B+\frac{p'_1}{p_1}(p'_2-p_2)} \end{aligned}$$

Therefore, if $q > \frac{c_A+c_B}{c_A+c_B+\frac{p'_1}{p_1}(p'_2-p_2)}$, A makes an aggressive demand $x_2 = p'_2 + c_B$ in the second stage. The weak type receives $1 - p'_1(p'_2 + c_B) - c_B$ after rejecting the first stage proposal. In this situation, the weak type has a profitable deviation. Had it instead accepted the offer, it would always get a payoff $1 - x_1$, which is better than its war payoff. We can easily prove it from the range of x_1 :

$$x_1 < p'_1(p'_2 + c_B) + c_B$$

¹¹Later, we will prove that when $\frac{p'_1}{p_1} < R^*$, the separating strategy dominates the semi-separating strategies only when $q < q_{U3}^*$. Since both $\frac{(c_A+c_B)}{\frac{p_1-1}{p_1}(c_A+c_B)+\frac{p'_1}{p_1}(p'_2-p_2)} > q_{U3}^*$ and $1 > q_{U3}^*$ always hold, the range of q for which the separating equilibrium occurs is ultimately $q \in [q_{U1}^*, q_{U3}^*)$.

$$1 - x_1 > 1 - p'_1(p'_2 + c_B) - c_B$$

If $q < \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, A has its posterior belief $r < r^*$ and makes a conservative demand $x_2 = p_2 + c_B$ in the second stage. The weak type receives $1 - p'_1(p_2 + c_B) - c_B$ after rejecting the first stage proposal. In this situation, the weak type has no profitable deviations since $1 - x_1 < 1 - p'_1(p_2 + c_B) - c_B$. However, A will always deviate to make the less risky demand $x_1 = p'_1(p_2 + c_B) + c_B$ in the first stage. For proof, A receives $q[p'_1(p_2 + c_B) - c_A] + (1 - q)[p'_1(p_2 + c_B) - c_A]$ when both types reject, which is worse than the separating strategy where A receives $q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p'_1(p_2 + c_B) - c_A]$.

Third, based on Lemma 7, both types accept proposals $x_1 \leq p_1(p_2 + c_B) + c_B$. In this situation, A receives its payoff under agreement x_1 . To optimize the utility, A demands $x_1 = p_1(p_2 + c_B) + c_B$.

Nevertheless, A will deviate to make a more aggressive demand when it holds a strong belief that B is a weak type. There are two situations that A may deviate. Based on Lemma 8, A makes an aggressive proposal $x_1 > p'_1(p'_2 + c_B) + c_B$ and there is a pooling strategy where both types reject. Given that B is sufficiently likely to be the weak type, the war payoff for A is still greater than what it can get from the agreement. Also, A makes a proposal in the moderately high range $x_1 \in (p'_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$, and there is a semi-separating strategy that the strong type rejects but the weak type mixes between accepting and rejecting.

If A proposes $x_1 > p'_1(p'_2 + c_B) + c_B$, A receives the war payoff based on its belief of the probability B is a weak type. Given that both types reject in the first stage, if A enters into the second stage, its posterior belief r equals the prior belief q . As is proved in the previous sections, if $q < \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, A proposes $x_2 = p_2 + c_B$ in the second stage and both types accept the offer. Then A will not deviate from the conservative proposal given the war payoff is always worse than the payoff for the separating strategy. If $q > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, however, A proposes $x_2 = p'_2 + c_B$ in the second stage. This time, A receives $p'_2 + c_B$ in the second stage when B is the weak type and accepts the offer and $p_2 - c_A$ when B is the strong type and rejects the offer. A's payoff when fighting a war with both types (indicating that both types adopt the pooling-on-fighting strategy) in the first stage becomes $q[p'_1(p'_2 + c_B) + c_B] + (1 - q)[p_1(p_2 - c_A) - c_A]$. To induce A to prefer the war demand ($x_1 > p'_1(p'_2 + c_B) + c_B$) over the conservative demand $p_1(p_2 + c_B) + c_B$, we have:

$$q[p'_1(p'_2 + c_B) - c_A] + (1 - q)[p_1(p_2 - c_A) - c_A] > p_1(p_2 + c_B) + c_B$$

$$q > \frac{(1 + p_1)(c_A + c_B)}{p'_1 p_2 - p_1 p_2 + p'_1 c_B + p_1 c_A}$$

However, we will later show that the pooling-on-fighting strategy will never be an equilibrium. When $\frac{p'_1}{p_1} > R^*$, $q_{U2}^* < \frac{(1+p_1)(c_A+c_B)}{p'_1 p_2 - p_1 p_2 + p'_1 c_B + p_1 c_A}$,¹² indicating that the pooling-on-fighting strategy is always dominated by the pooling-on-accepting strategy when $q \leq q_{U2}^*$. In the next section, we will prove that, as long as $\frac{p'_1}{p_1} > R^*$ holds, the pooling-on-fighting strategy is also dominated by the semi-separating strategy when $q > q_{U2}^*$.

When $\frac{p'_1}{p_1} < R^*$, $q_{U1}^* < \frac{(1+p_1)(c_A+c_B)}{p'_1 p_2 - p_1 p_2 + p'_1 c_B + p_1 c_A}$, implying that the pooling-on-fighting strategy is always dominated by the pooling-on-accepting strategy when $q \leq q_{U1}^*$. In the next section, we will show that, as long as $\frac{p'_1}{p_1} < R^*$ holds, the pooling-on-fighting strategy is either dominated by the separating strategy or by the semi-separating strategy when $q > q_{U1}^*$.

A.5.4 Proof of Proposition 11

First, when the proposals are in the moderately high range $x_1 \in (p'_1(p_2 + c_B) + c_B, p'_1(p'_2 + c_B) + c_B]$, the strong type always reject in the first stage. In a semi-separating equilibrium, the weak type mixes between accepting and rejecting in the first stage, and A is indifferent between making the conservative demand and the aggressive demand in the second stage. Let σ_R be the probability the weak type reject in the first stage, and $1 - \sigma_R$ be the probability the weak type accept. By Bayes' Rule, we can have A's posterior belief that B is the weak type as a function of q , σ_R and the first-stage power distribution p_1 and p'_1 : $\frac{q\sigma_R p'_1}{q\sigma_R p'_1 + (1-q)p_1}$. As is proved before, A is indifferent between the two proposals in the second stage if $r = r^*$. Solving for σ_R :

$$\frac{q\sigma_R p'_1}{q\sigma_R p'_1 + (1-q)p_1} = \frac{c_A + c_B}{c_A + c_B + p'_2 - p_2}$$

$$\sigma_R = \left(\frac{1-q}{q} \right) \left(\frac{c_A + c_B}{p'_2 - p_2} \right) \frac{p_1}{p'_1}$$

¹²For proof, note that $q_{U1}^* < \frac{(1+p_1)(c_A+c_B)}{p'_1 p_2 - p_1 p_2 + p'_1 c_B + p_1 c_A}$ always holds, since $\frac{(1+p_1)(c_A+c_B)}{p'_1 p_2 - p_1 p_2 + p'_1 c_B + p_1 c_A} = \frac{(1+p_1)(c_A+c_B)}{p_1(c_A+c_B) + (p_2+c_B)(p'_1-p_1)} > q_{U1}^* \equiv \frac{c_A+c_B}{c_A+c_B + (p_2+c_B)(p'_1-p_1)}$. Later, we will prove that when $\frac{p'_1}{p_1} > R^*$, $q_{U2}^* < q_{U1}^*$. Therefore, under this condition, $q_{U2}^* < \frac{(1+p_1)(c_A+c_B)}{p'_1 p_2 - p_1 p_2 + p'_1 c_B + p_1 c_A}$.

Second, let $\sigma_{Rp}^* \equiv \left(\frac{1-q}{q}\right) \left(\frac{c_A+c_B}{p'_2-p_2}\right) \frac{p_1}{p'_1}$.¹³ Then we have A's payoff in the first stage as follows:

$$q(1 - \sigma_{Rp}^*)x_1 + q\sigma_{Rp}^*[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A]$$

To optimize its payoff, A will set x_1 to its largest feasible value $p'_1(p'_2 + c_B) + c_B$. Ultimately, A's payoff becomes:

$$q(1 - \sigma_{Rp}^*)[p'_1(p'_2 + c_B) + c_B] + q\sigma_{Rp}^*[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A]$$

Finally, the semi-separating equilibrium exists only when A has no profitable deviation. To make A prefer the semi-separation demand $x_1 = p'_1(p'_2 + c_B) + c_B$ to the most conservative one $x_1 = p_1(p_2 + c_B) + c_B$, we have:

$$q(1 - \sigma_{Rp}^*)[p'_1(p'_2 + c_B) + c_B] + q\sigma_{Rp}^*[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A] > p_1(p_2 + c_B) + c_B$$

$$q > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1(p'_2 - p_2) + (p_2 + c_B)(p'_1 - p_1)}{p_1 + 1 + \frac{p'_1}{p_1} \left(\frac{c_A + c_B}{p'_2 - p_2}\right)}} \equiv q_{U2}^*$$

We also need to ensure that A prefers the semi-separation demand $x_1 = p'_1(p'_2 + c_B) + c_B$ to the war demand $x_1 > p'_1(p'_2 + c_B) + c_B$. If $q < \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, A always makes the conservative demand in the second stage, the war payoff $q[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A]$ is dominated by the semi-separating equilibrium. When $q > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$, A makes the aggressive demand in the second stage, which is also the condition that the pooling-on-rejecting strategy is dominated by the semi-separating strategy:

$$q(1 - \sigma_{Rp}^*)[p'_1(p'_2 + c_B) + c_B] + q\sigma_{Rp}^*[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A] >$$

$$q[p'_1(p'_2 + c_B) - c_A] + (1 - q)[p_1(p_2 - c_A) - c_A]$$

$$q > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$$

¹³ $\sigma_{Rp}^* \in (0, 1]$ when $q \geq \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$. For proof, σ_{Rp}^* is always larger than 0. To make σ_{Rp}^* less than or equal to 1, we have $q \geq \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$.

Finally, A should also prefer the semi-separation demand $x_1 = p'_1(p'_2 + c_B) + c_B$ to the moderately aggressive demand $x_1 = p'_1(p_2 + c_B) + c_B$ where the separating strategy exists. To ensure no deviation, we have:

$$q(1 - \sigma_{Rp}^*)[p'_1(p'_2 + c_B) + c_B] + q\sigma_{Rp}^*[p'_1(p_2 + c_B) - c_A] + (1 - q)[p_1(p_2 + c_B) - c_A] >$$

$$q[p'_1(p_2 + c_B) + c_B] + (1 - q)[p_1(p_2 + c_B) - c_A]$$

$$q > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1(p'_2 - p_2)}{p_1 + \frac{p_1}{p'_1} \left(\frac{c_A + c_B}{p'_2 - p_2} \right)}} \equiv q_{U3}^*$$

If $\frac{p'_1}{p_1} < \left(\frac{p'_1 - p_1}{p'_2 - p_2} \right) (p_2 + c_B) \left[1 + \frac{(c_A + c_B)}{p'_1(p'_2 - p_2)} \right]$, $q_{U3}^* > q_{U2}^* > q_{U1}^*$. Therefore, when $q < q_{U1}^*$, both types accept (**Proposition 12: Pooling**); when $q \in [q_{U1}^*, q_{U3}^*]$, the strong type rejects while the weak type accepts (**Proposition 10: Separating**); when $q > q_{U3}^*$, the strong type always rejects, while the weak type rejects with a probability σ_{Rp}^* (**Proposition 11: Semi-Separating**).

If $\frac{p'_1}{p_1} > \left(\frac{p'_1 - p_1}{p'_2 - p_2} \right) (p_2 + c_B) \left[1 + \frac{(c_A + c_B)}{p'_1(p'_2 - p_2)} \right]$, $q_{U3}^* < q_{U2}^* < q_{U1}^*$.¹⁴ This time, however, the separating equilibrium doesn't exist. When $q < q_{U2}^*$, both types accept (**Proposition 12: Pooling**); when $q > q_{U2}^*$, the strong type always rejects, while the weak type rejects with a probability σ_{Rp}^* (**Proposition 11: Semi-Separating**).

In evaluating the relationship between $\frac{p'_1}{p_1}$ and $\left(\frac{p'_1 - p_1}{p'_2 - p_2} \right) (p_2 + c_B) \left[1 + \frac{(c_A + c_B)}{p'_1(p'_2 - p_2)} \right]$, the separating equilibrium exists only when p_1 is at the middle level. For proof, if p_1 converges to 0, the left hand side $\frac{p'_1}{p_1}$ becomes extremely huge. If p_1 converges to its highest feasible value p'_1 , the right hand side $\left(\frac{p'_1 - p_1}{p'_2 - p_2} \right) (p_2 + c_B) \left[1 + \frac{(c_A + c_B)}{p'_1(p'_2 - p_2)} \right]$ converges to 0. For both situation, $\frac{p'_1}{p_1} < R^*$ fails to hold.

A.6 Additional Comparative Statics

In addition to the comparative statics shown in Figure 3, we also examine two alternative comparative statics to better illustrate the equilibrium of the baseline model. First, we plot how the probability of war changes with the gap in second-stage capacities between the strong and weak types, $p'_2 - p_2$, holding constant total battle

¹⁴Once $\frac{p'_1}{p_1} > R^*$ holds, we also have $q_{U2}^* > \frac{c_A + c_B}{c_A + c_B + \frac{p'_1}{p_1}(p'_2 - p_2)}$. Therefore, $q > q_{U2}^*$ is the condition for the semi-separating equilibrium to exist.

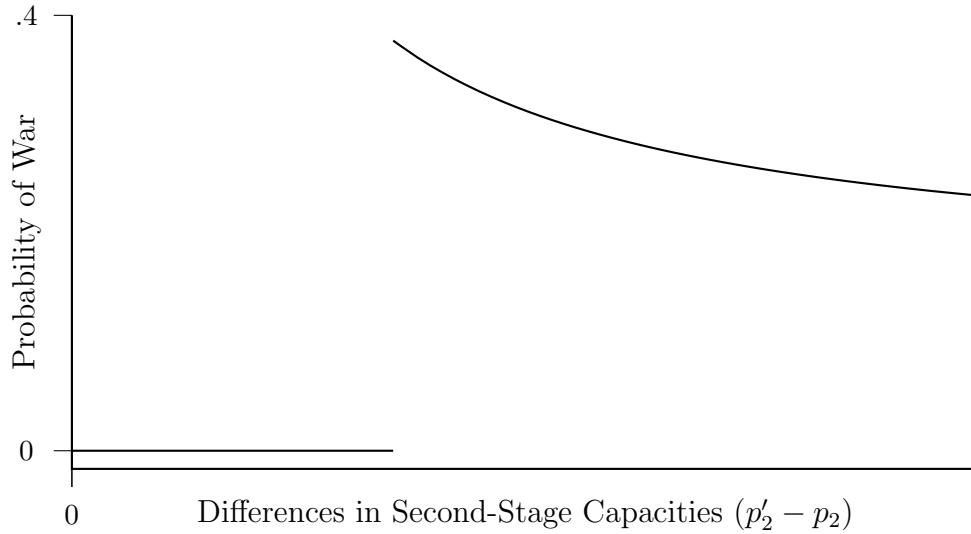


Figure A.5: The Differences in Second-Stage Capacities ($p'_2 - p_2$) and the Probability of War

inefficiency and the prior belief about B's type. Figure A.5 illustrates this relationship. When the gap between the weak and strong types is small, the equilibrium is a pooling strategy in which A makes a conservative demand that appeases both types, driving the probability of war to zero. If the two types are identical, the model converges to a complete-information game in which no war occurs. As the gap between the two types becomes sufficiently large, the equilibrium shifts to a semi-separating strategy: the strong type always rejects and fights in the first stage, while the weak type rejects with some probability. However, as the gap widens further, screening becomes more effective. The weak type becomes increasingly likely to accept the offer under the semi-separating equilibrium, thereby reducing the overall probability of war.

Second, we plot the relationship between A's first-stage demand, x_1 , and actor B's battle inefficiency in Figure A.6, holding c_A constant. In general, an increase in B's cost of war expands the range of demands that A can profitably make in the first stage. However, the demand drops sharply once battle inefficiency reaches a threshold that induces A to shift from an aggressive demand that extracts concessions from the weak type to a more conservative demand that appeases both types, and then increases monotonically as battle inefficiency continues to rise.

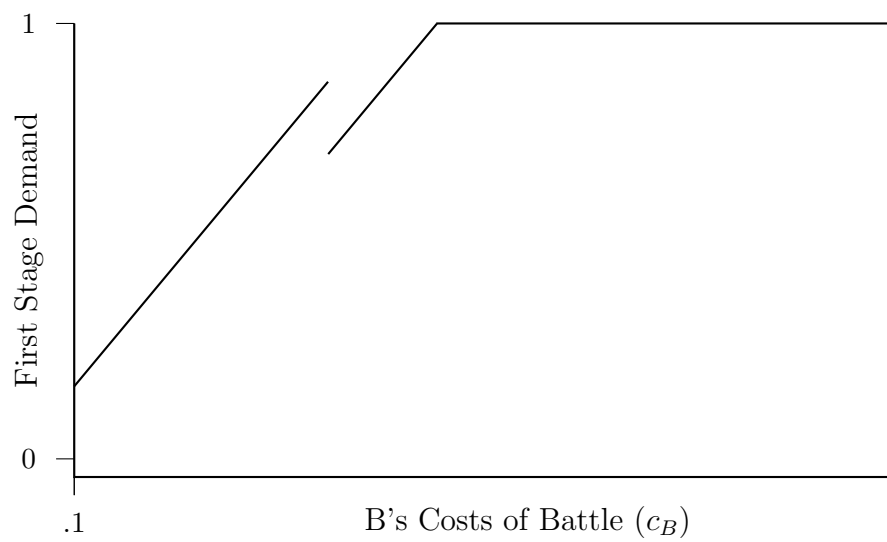


Figure A.6: B's Inefficiency in Battle (c_B) and First-Stage Demands